10 Z – Transform

Objective

After completing this chapter, the user will be able to understand

1. Mapping between s and z-plane.
2. Bilateral Z-transform and concept of ROC.
3. Properties of bilateral Z-transform and ROC.
4. Unilateral Z-transform and its properties.
5. Solution of difference equation.Block diagram representation.

Introduction

1. Z-transform, which is the discrete-time counterpart of the Laplace transform. Just as the Laplace transform provides us with a frequency-domain technique for analysing signals for which the CTFT does not exist, the z-transform enables us to analyse certain discrete-time signals that do not have a discrete-time Fourier transform (DTFT).
2. The z-transform comes into two varieties: Bilateral or two-sided z-transform and unilateral, or one-sided z-transform.
4. The bilateral z-transform offers insight into the nature of system characteristics, such as stability, causality, and frequency response.
5. The unilateral z-transform is a convenient tool for solving differential equations with initial conditions.

Right Sided Sequence: (RSS) $\rightarrow$ Duration of $x(n) \rightarrow \infty, x(n) = 0$ toward the left of some $n = n_0$.

Left Side Sequence: (LSS) $\rightarrow$ Duration of $x(n) \rightarrow \infty, x(n) = 0$ towards the right of some $n = n_0$.

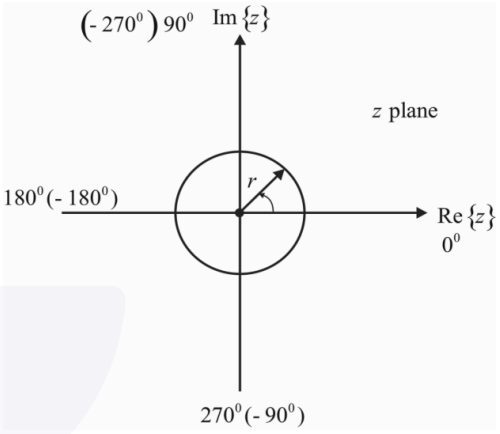
Both sided sequence: Duration of $x(n) \rightarrow \infty$ $x(n) \rightarrow$ L.S.S. + R.S.S. around $n = n_0$

Finite duration +Absolutely summable:
Z-plane

1. Polar complex plane
2. $z = re^{j\omega}$

   $|z| = r \rightarrow 0 \le r \le \infty$

   $\angle z = \omega \rightarrow 0 < \omega \le 2\pi$



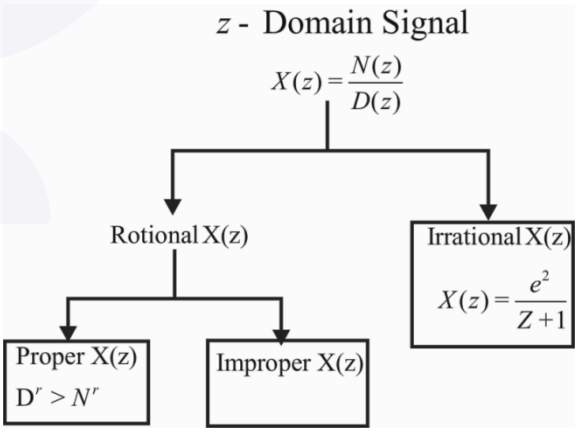


Fig. 10.1

Poles and zeros of Z-domain signal:

Pole: $z = z_p \quad \lim_{z \to z_p} X(z) \rightarrow \pm\infty$

Zero: $z = z_z : \quad \lim_{z \to z_z} X(z) = 0$

If $\rightarrow X(z)$: Rational

1. Common factor cancel
2. Root of $D(z)$: poles

Pole mapping on Z-plane:

$$X(z) = \frac{1}{(2 - 2e^{j30^0})}$$

Pole: $z = 2e^{j30^0}$ (point)

$|z| = 2$ (circle)

$|z| < 2$ (disc): 2D Region

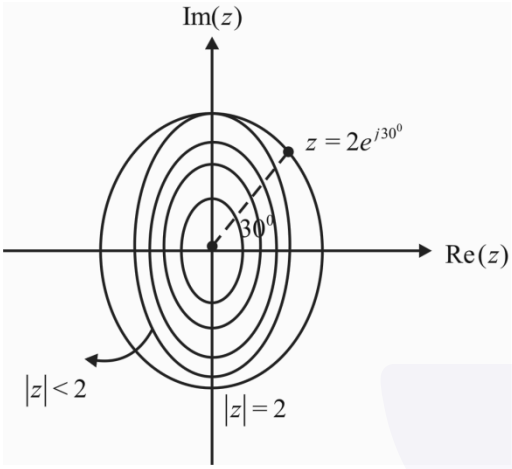


Fig. 10.2

$z = e^{sT_s}$

$nT_s \to n$

$$Y(s) = \sum_{n=-\infty}^{\infty} x(n)z^{-n} = Y(z)$$

Note:

1. Transformation parameter: $z = e^{sT_s}$
2. $x(t) \longleftrightarrow X(s)$

   $\updownarrow$ sample $\updownarrow$ $s = \frac{\ln z}{T_s}$

   $x(n) \xleftrightarrow[\text{Formula}]{Z.T.} X(z) = ?$

3. Journey from $x(s)$ to $x(z)$

$$X(s) \xrightarrow{ILT} x(t) \xrightarrow{sampling} x(n) \xrightarrow{Z.T.} x(z)$$

$z = e^{sT_s} = e^{\sigma T_s} e^{j\omega T_s}$

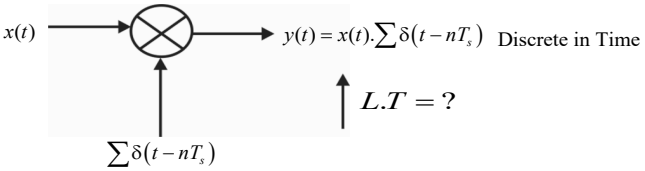
$|z| = e^{\sigma T_s}$

Mapping between S -plane and Z-plane:

Laplace Transform	Z -Transform
$s = \sigma + j\omega$	$z = re^{t\omega}$
$s = a + jb$: point	$z = r_0 e^{j\omega_0}$: point
$\text{Re}\lvert s\rvert = 0$: Line parallel to $j\omega$ axis	$\lvert z\rvert = r_0$: Circle concentric to unity radius circle $\lvert z\rvert = 1$
$\text{Re}\lvert s\rvert > a$: Region parallel to $j\omega$ axis	$\lvert z\rvert > r_0$: Region concentric to unity radius circle.

Table 10.1

$x(t) \longleftrightarrow x(s)$



$$y(t) = \sum_{n=-\infty}^{\infty} x(nTs)\delta(t - nT_s)$$

$$Y(s) = \sum_{n=-\infty}^{\infty} x(nT_s)e^{-nT_s(s)}$$

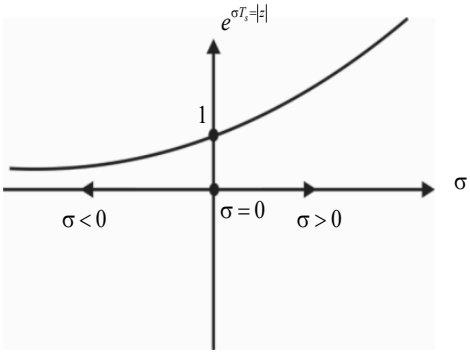


Fig. 10.3

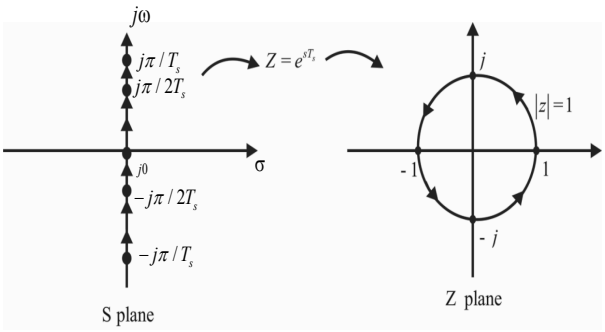


Fig. 10.4

Case a)

$\sigma < 0$: Left half side of a plane

$\updownarrow$

$0 \le |z| < 1$: Inside unity circle

Case b)

$\sigma = 0$: $j\omega$ axis

$\updownarrow$

$|z| = 1$: Unity circle itself

Case c)

$\sigma > 0$: Right half side of s plane

$\updownarrow$

$|z| > 1$: Outside unity circle

$z = e^{sT_s}$

$s = \dfrac{-j\pi}{T_s} \longrightarrow z = -1 \qquad |z| = 1$

$s = \dfrac{-j\pi}{2T_s} \longrightarrow z = -j \qquad |z| = 1$

$s = jo \longrightarrow z = 1 \qquad |z| = 1$

$s = \dfrac{j\pi}{2T_s} \longrightarrow z = j \qquad |z| = 1$

$s = \dfrac{j\pi}{Ts} \longrightarrow z = -1 \qquad |z| = 1$

Important points of mapping:

1. Each vertical line in s-plane $\left(\dfrac{-j\pi}{T_s} \text{ to } \dfrac{j\pi}{T_s}\right)$ is mapped to a circle in ACW in Z-plane.
2. Left half side of s plane is mapped inside the unity circle in Z-plane.
3. Left sided nature of s plane is converted into inward nature of Z-plane.
4. Right half side of s plane is mapped outside the unity circle.
5. Right sided nature of s plane is converted into outward nature in Z-plane.
6. $j\omega$ axis is mapped onto unity circle.
7. The origin of s plane is mapped $Z = e^{sT_s} = 1$ $(0,0) \rightarrow \sigma = 0 \quad \omega = 0 \quad s = 0$.

Important analogy:

$1 = u(t) + u(-t)$

$(1)^n = u(n) + u(-n-1)$

C.T.signal		D.T.signal
$u(t)$	$\longleftrightarrow$	$u[n]$
$u(-t)$	$\longleftrightarrow$	$u(-n-1)$
$e^{-\alpha t}u(t)$	$\longleftrightarrow$	$a^n u[n]$
$-e^{-\alpha t}u(-t)$	$\longleftrightarrow$	$-a^n u[-n-1]$

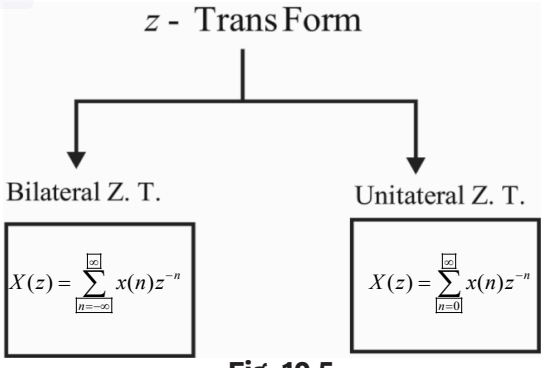


Fig. 10.5

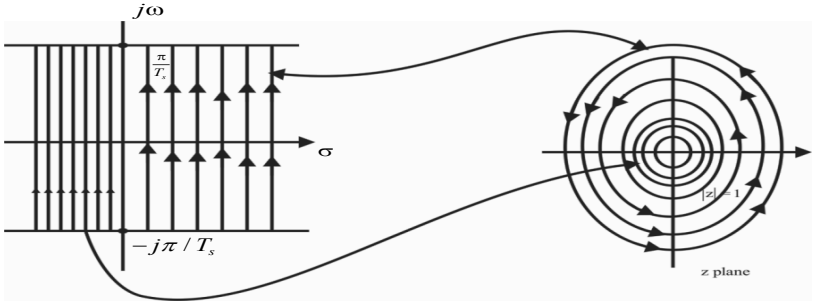


Fig. 10.6

1. $x(n) = (z_0)^n u(n)$

$z_0 = r_0 e^{j\omega_0}$: complex constant

Solution:

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$X(z) = \sum_{n=-\infty}^{\infty} (z_0)^n u(n) z^{-n}$$

$$X(z) = \sum_{n=-\infty}^{\infty} (z_0^{-1} z)^{-n}$$

$$= 1 + \left(\frac{z}{z_0}\right)^{-1} + \left(\frac{z}{z_0}\right)^{-2} + \left(\frac{z}{z_0}\right)^{-3} + \ldots$$

$$X(z) = 1 + \underbrace{\left(\frac{z}{z_0}\right)^{-1}}_{r} + \left(\frac{z}{z_0}\right)^{-2} + \ldots$$

$$x(z) = \begin{cases} \dfrac{1}{1-(z/z_0)^{-1}} & \text{when } \left|\left(\dfrac{z}{z_0}\right)^{-1}\right| < 1 \rightarrow \left|\dfrac{z_0}{z}\right| < 1 \\ \infty & \text{otherwise} \end{cases}$$

$$\frac{|z_0|}{|z|} < 1$$

$$x(z) = \begin{cases} \dfrac{z}{z-z_0} & : \ |z| > |z_0| \\ \infty & : \text{otherwise} \end{cases}$$

Region in which x(z) converges (ROC)

$$x(n) = (z_0)^n u(n) \longleftrightarrow X(z) = \frac{z}{z-z_0} : |z| > |z_0|$$

$$x(n) = (z_0)^n u(n) \longleftrightarrow X(z) = \frac{z}{z-z_0}$$

pole: $z = z_0$

$|z| = |z_0|$

$|z| > |z_0|$: outward:

$ROC: \ |z_0| < |z| \le \infty$

$$(z_0)^n u(n) = \left\{ \underset{\uparrow}{a} \quad b \quad c \quad d \right\} \underset{n=0}{} \longleftrightarrow X(z) =$$

$a + bz^{-1} + cz^{-2} + dz^{-3}$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} =$$

$\ldots x(-2)z^2 + x(-1)z^1 + x(0) + x(1)z^{-1} + x(2)z^{-2}$

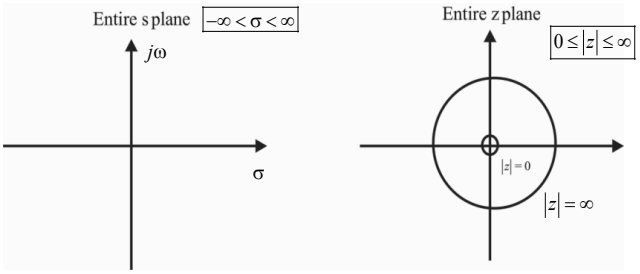


Fig. 10.7

2. $x(n) = -z_0^n u(-n-1)$

Solution:

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} = -\sum_{n=-\infty}^{-1} z_0^n z^{-n}$$

$$= -\sum_{n=-1}^{\infty} z_0^n z^{-n} = -\sum_{n=-1}^{-\infty} \left(\frac{z_0}{z}\right)^n$$

$$X(z) = -\left[\left(\frac{z_0}{z}\right)^{-1} + \left(\frac{z_0}{z}\right)^{-2} + \ldots\infty\right]$$

$$X(z) = \frac{-\left(\dfrac{z_0}{z}\right)^{-1}}{1-\left(\dfrac{z_0}{z}\right)^{-1}} \quad \text{when} \quad \left|\left(\frac{z_0}{z}\right)^{-1}\right| < 1$$

$$X(z) = \begin{cases} \dfrac{z}{z-z_0} & : \ |z| < |z_0| \\ \infty & : \text{otherwise} \end{cases}$$

$$-z_0^n \overbrace{u(-n-1)}^{-\infty \le n \le -1} \longleftrightarrow X(z) = \frac{z}{z-z_0}$$

Pole: $z = z_0$

$|z| = |z_0|$

$|z| < |z_0|$

ROC: $0 \le |z| < |z_0|$

3. $x(n) = \delta(n) = \{\underset{\uparrow}{1}\}$

Solution:

$$X(z) = \sum_{n=-\infty}^{\infty} \delta(\underbrace{n}_{n=0}) z^{-n} = \sum_{n=-\infty}^{\infty} \delta(n) = 1$$

$\delta(n) \longrightarrow 1$ ROC: Entire $z-$plane

$0 \le |z| \le \infty$

4. $x(n) = \left\{\underset{\uparrow}{1},\ 2,\ 3\right\}$

Solution:

$$x(n) = \left\{\underset{\uparrow}{1},\ 2,\ 3\right\}_{n=0}$$

$X(z) = 1 + 2z^{-1} + 3z^{-2}$ ROC: Entire z-plane except

$0 < |z| \le \infty$ $|z| = 0$

5. $x(n) = \left\{1,\ 2,\ \underset{\uparrow}{3}\right\}$

Solution:

$$x(n) = \underset{n=-2}{\underset{\uparrow}{1}},\ \underset{n=-1}{2},\ \underset{n=0}{3}$$

$X(z) = 3 + 2z + z^2$

$0 \le |z| < \infty$ ROC: entire z-plane except $|z| = \infty$

6. $x(n) = \{1,\ \underset{\uparrow}{2},\ 3\}$

Solution:

$$x(n) = \underset{-1}{1},\ \underset{n=0}{\underset{\uparrow}{2}},\ \underset{1}{3}$$

$X(z) = z + 2 + 3z^{-1}$ $0 < |z| < \infty$: ROC : entire z plane except $|z| = 0,\ |z| = \infty$

Points To Remember

1. $a^n u(n) \longleftrightarrow \dfrac{z}{z-a}$ $ROC : |z| > |a|$
2. $-a^n u(-n-1) \longleftrightarrow \dfrac{z}{z-a}$ $ROC : |z| < |a|$
3. $a^{-n} u(n) \longleftrightarrow \dfrac{z}{z-1/a}$ $ROC : |z| > \dfrac{1}{|a|}$
4. $-a^{-n} u(-n-1) \longleftrightarrow \dfrac{z}{z-1/a}$ $ROC : |z| < \dfrac{1}{|a|}$
5. $(-a)^{-n} u(n) \longleftrightarrow \dfrac{z}{z+a}$ $ROC : |z| > |a|$
6. $-(-a)^n u(-n-1) \longleftrightarrow \dfrac{z}{z+a}$ $ROC : |z| < |a|$
7. $(-a)^{-n} u(n) \longleftrightarrow \dfrac{z}{z+\frac{1}{a}}$ $|z| > \dfrac{1}{|a|}$
8. $-(-a)^{-n} u(-n-1) \longleftrightarrow \dfrac{z}{z+\frac{1}{a}}$ $|z| < \dfrac{1}{|a|}$
9. $u(n) \longleftrightarrow \dfrac{z}{z-1}$ $|z| > 1$
10. $-u(-n-1) \longleftrightarrow \dfrac{z}{z-1}$ $|z| < 1$

7. $x(n) = -(1-3j)^n u(-n-1)$

Solution:

$$x(n) = -(1-3j)^n \underbrace{u(-n-1)}_{-\infty \le n \le -1} \longleftrightarrow X(z)$$

$$= \frac{z}{z-(1-3j)}$$

$pole$: $z = 1-3j$

$|z| = \sqrt{10}$

$|z| < \sqrt{10}$

$0 \le |z| < \sqrt{10}$

Properties of BZT:

Linearity: $x_1(n) \longleftrightarrow X_1(z)$ $ROC : R_1$

$x_2(n) \longleftrightarrow X_2(z)$ $ROC : R_2$

Case 1:

$x(n) = Ax_1(n) + Bx_2(n) \longleftrightarrow R_1 \cap R_2 = \{\phi\}$ Z.T. DNE

R•S•S• $R_1 \cap R_2 \pm \{\phi\}$

L•S•S•

B•S•S• $X(z) = AX_1(z) + BX_2(z)$: $R = R \cap R_2$

Case 2:

$Ax_1(n) + Bx_2(n) \longleftrightarrow AX_1(z) + BX_2(z)$

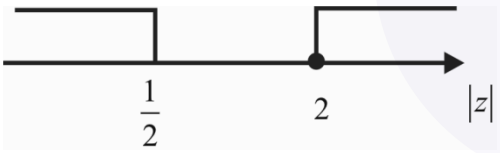
$\rightarrow F \cdot D \cdot + Abs \sum$ ROC: entire z-plane except

Possibly $|z| = 0$ And/Or $|z| = \infty$

8. $x(n) = 2^{|n|} \longleftrightarrow X(z) = ?$

Solution:

$$x(n) = \underbrace{2^n u(n)}_{R\bullet S\bullet S} + \underbrace{2^{-n} u(-n-1)}_{L\bullet S\bullet S}$$



$2^n u(n) \longleftrightarrow \frac{z}{z-2}$ $|z| > |2|$

$\left(\frac{1}{2}\right)^n u(-n-1) \longleftrightarrow \frac{-z}{z-\frac{1}{2}}$ $|z| < \left|\frac{1}{2}\right|$

$\underbrace{x(n)}_{B\bullet S\bullet S\bullet} \longleftrightarrow R_1 \cap R_2 = \{\phi\}$ Z.T. D•N•E•

Test Your Understanding

1. Find the ZT of the following signal.

$$x(n) = \left(\frac{1}{2}\right)^{|n|}$$

Solution:

$\underbrace{x(n)}_{B\bullet S\bullet S\bullet} \overset{Z.T}{\longleftrightarrow} R_1 \cap R_2$:

$\frac{1}{2} < |z|, 2$: $\frac{z}{z-\frac{1}{2}} - \frac{z}{z-2}$

↓ ROC

2. $x(n) = (1-3j)^{-|n|}$

Solution:

$$X(z) = \frac{z}{z-\frac{1}{1-3j}} - \frac{z}{z-(1-3j)}$$

Time shifting:

$x(n) \longleftrightarrow X(z)$ ROC: R

$x(n-n_0) \longleftrightarrow z^{-n_0} X(z)$ ROC: R

$x(n-n_0) \longleftrightarrow z^{n_0} X(z)$ except possible inclusion/removal of $|z| = 0$ and/or $|z| = \infty$

9.

$\underbrace{u(n)}_{0\le N<\infty} \longleftrightarrow \frac{z}{z-1}$ ROC : $1 < |Z| \le \infty$

$\underbrace{u(n+1)}_{-1\le n<\infty} \longleftrightarrow z\left\{\frac{z}{z-1}\right\}$ ROC : $1 < |Z| < \infty$

$$\underbrace{u(n-1)}_{1\le n<\infty} \longleftrightarrow z^{-1}\left\{\frac{z}{z-1}\right\} \quad ROC \;:\; 1<|Z|\le\infty$$

10. $u(-n-1) \longleftrightarrow \dfrac{-z}{z-1}$ $\quad ROC \;:\; 0\le|z|<1$

$$u(-(n-1)-1)=\underbrace{u(-n)}_{-\infty<n\le 0} \longleftrightarrow z^{-1}\left\{\frac{-z}{z-1}\right\} \quad :\quad 0\le|z|<1$$

$$u(-(n-2)-1)=\underbrace{u(-n+1)}_{-\infty<n\le 1} \longleftrightarrow z^{-2}\left\{\frac{-z}{z-1}\right\} \quad :\quad 0\le|z|<1$$

Multiplication with complex exponential:

$x(n) \longleftrightarrow X(z)$ $\quad ROC : |z|$ $\qquad$ $z_{0:}$ complex constant

$$(z_0)^n x(n) \longleftrightarrow X\left(\frac{z}{z_0}\right) \quad ROC : \left|\frac{z}{z_0}\right|$$

Extension:

$x(n) \longleftrightarrow X(z)$ $\quad ROC : |z|$

$$(-1)^n x(n) \longleftrightarrow X(-z) \quad ROC : \left|\frac{z}{-1}\right| = |z|$$

11. $2^n u(n+3) \xleftrightarrow{Z.T} ?$

Solution:

$u(n) \longleftrightarrow \dfrac{z}{z-1}\, ROC : |z|>1$ $\qquad$ $u(n) \longleftrightarrow \dfrac{z}{z-1}$

$u(n+3) \longleftrightarrow \dfrac{z^4}{z-1}\, ROC : |z|>1$ $\qquad$ $u(n+3) \longleftrightarrow \dfrac{z^4}{z-1}$

$$2^n u(n+3) \longleftrightarrow \frac{\left(\frac{z}{2}\right)^4}{\left(\frac{z}{2}\right)-1}\, ROC : \left|\frac{z}{2}\right|>1 \qquad \underbrace{2^n u(n+3)}_{RSS} \longleftrightarrow \frac{\left(\frac{z}{2}\right)^4}{\left(\frac{z}{2}\right)-1}$$

$|z|>2$

$z=2$
$|z|=2$
$|z|>2$

1. $\longrightarrow u(n) \longleftrightarrow \dfrac{z}{z-1}$ $\qquad ROC \quad |z|>1$

2. $\longrightarrow e^{j\omega_0 n}u(n) \longleftrightarrow \dfrac{\dfrac{z}{e^{j\omega_0}}}{\dfrac{z}{e^{j\omega_0}}-1}$ $\quad ROC \quad \left|\dfrac{z}{e^{j\omega_0}}\right|>1 \quad : |z|>1$

3. $\longrightarrow e^{-j\omega_0 n}u(n) \longleftrightarrow \dfrac{z(e^{j\omega_0})}{ze^{j\omega_0}-1}$ $\quad ROC \quad : \left|ze^{j\omega_0}\right|>1 \quad : |z|>1$

a) $\sin \omega_0 nu(n) \longleftrightarrow \dfrac{z\sin\omega_0}{z^2-2z\cos\omega_0+1} : |z|>1$

b) $\cos \omega_0 nu(n) \longleftrightarrow \dfrac{z^2-z\cos\omega_0}{z^2-2z\cos\omega_0+1} : |z|>1$

c) $a^n \sin \omega_0 nu(n) \longleftrightarrow \dfrac{az\sin\omega_0}{z^2-2az\cos\omega_0+a^2} : |z|>|a|$

d) $a^n \cos \omega_0 nu(n) \longleftrightarrow \dfrac{z^2-az\cos\omega_0}{z^2-2az\cos\omega_0+a^2} : |z|>|a|$

e) $-\sin \omega_0 nu(-n-1) \longleftrightarrow \dfrac{z\sin\omega_0}{z^2-2z\cos\omega_0+1} : |z|<1$

f) $-\cos \omega_0 nu(-n-1) \longleftrightarrow \dfrac{z^2-z\cos\omega_0}{z^2-2z\cos\omega_0+1} : |z|<1$

g) $-a^n \sin \omega_0 u(n-1) \longleftrightarrow \dfrac{az\sin\omega_0}{z^2-2az\omega\delta gw_0+a^2} \; |z|<|a|$

h) $-a^n \cos \omega_0 nu(-n-1) \longleftrightarrow \dfrac{z^2-az\cos\omega_0}{z^2-2az\cos\omega_0+a^2} |z|<|a|$

Time reversal:

$x(n) \longleftrightarrow X(z) : ROC : |z|$

$x(-n) \longleftrightarrow X(z^{-1}) : ROC : |z^{-1}|$

1. $u(n) \longleftrightarrow \dfrac{z}{z-1} \qquad |z|>1$

2. $-u(n) \longleftrightarrow \dfrac{-z}{z-1} \qquad |z|>1$

3. $u(-n) \longleftrightarrow \left\{\dfrac{z^{-1}}{z^{-1}-1}\right\} \qquad |z^{-1}|>1 \rightarrow |z|<1$

4. $(-1)^n u(n) \longleftrightarrow \left\{\dfrac{\dfrac{z}{(-1)}}{\dfrac{z}{(-1)}-1}\right\} \quad \left|\dfrac{z}{-1}\right| > 1 \rightarrow |z| > 1$

5. $-u(-n-1) \longleftrightarrow \dfrac{z}{z-1} \quad |z| < 1$

6. $u(-n-1) \longleftrightarrow \dfrac{-z}{z-1} \quad |z| < 1$

7. $-u(n-1) \longleftrightarrow \dfrac{z^{-1}}{z^{-1}-1} \quad |z^{-1}| < 1 \longrightarrow |z| > 1$

8. $-(-1)^n u(-n-1) \longleftrightarrow \dfrac{\dfrac{z}{-1}}{\dfrac{z}{-1}-1} \quad \left|\dfrac{z}{-1}\right| < 1$

Time scaling:

$x(n) \longleftrightarrow X(z) \quad ROC : |z|$

$x\left(\underset{K\in I}{\dfrac{n}{k}}\right) \longleftrightarrow X(z^k) \quad ROC : |z^k|$

12. $x(n) = 2^{\frac{n}{3}} u\left(\dfrac{n}{3}\right)$

Solution:

$x(n) = 2^{\frac{n}{3}} u\left(\dfrac{n}{3}\right)$

$2^n u(n) \longleftrightarrow \dfrac{z}{z-2} \quad Roc : |z| > 2$

$2^{\frac{n}{3}} u\left(\dfrac{n}{3}\right) \longleftrightarrow \underset{\substack{z=2^{\frac{1}{3}} \\ |z|=2^{\frac{1}{3}} \\ |z|>2^{\frac{1}{3}}}}{\dfrac{z^3}{z^3-2}} \quad ROC : \underset{|z|>(2)^{\frac{1}{3}}}{|z|^3 > 2}$

Multiplication by n in time domain property:

$x(n) \longleftrightarrow X(z) \quad ROC : |z|$

$nx(n) \longleftrightarrow -Z\dfrac{dx(z)}{dz} = x_1(z) \quad ROC : |z|$

$n^2 x(n) \longleftrightarrow -Z\dfrac{dx_1(z)}{dz} \quad ROC : |z|$

Important points:

1. $x(n) \longleftrightarrow X(z)$

$nx(n) \longleftrightarrow -z\dfrac{dx(z)}{dz}$

$(n-1)x(n-1) \longleftrightarrow \dfrac{-dX(z)}{dz}$

$(1-n)x(n-1) \longleftrightarrow \dfrac{dx(z)}{dz}$

2. $a^n u(n) \longleftrightarrow \dfrac{z}{z-a} \quad |z| > |a|$

$na^n u(n) \longleftrightarrow \dfrac{az}{(z-a)^2} \quad |z| > |a|$

$-a^n u(-n-1) \longleftrightarrow \dfrac{z}{z-a} \quad |z| < |a|$

$-na^n u(-n-1) \longleftrightarrow \dfrac{az}{(z-a)^2} \quad |z| < |a|$

3. $a^n u(n) \longleftrightarrow \dfrac{z}{z-a} \quad |z| > |a|$

$na^{n-1} u(n) \longleftrightarrow \dfrac{z}{(z-a)^2} \quad |z| > |a|$

$\dfrac{n(n-1)a^{n-2}}{2!} u(n) \longleftrightarrow \dfrac{z}{(z-a)^3} \quad |z| > |a|$

$\dfrac{n(n-1)(n-2)a^{n-3}}{3!} u(n) \longleftrightarrow \dfrac{z}{(z-a)^4} \quad |z| > |a|$

4. $-a^n u(-n-1) \longleftrightarrow \dfrac{z}{z-a} \quad |z| < |a|$

$\dfrac{-na^{n-1}}{1!} u(-n-1) \longleftrightarrow \dfrac{z}{(z-a)^2} \quad |z| < |a|$

$\dfrac{-n(n-1)a^{n-2}}{2!} u(-n-1) \longleftrightarrow \dfrac{z}{(z-a)^3} \quad |z| < |a|$

$\dfrac{-n(n-1)(n-2)a^{n-3}u(-n-1)}{3!} \longleftrightarrow \dfrac{z}{(z-a)^4} \quad |z| < |a|$

5. $na^n u(n) \longleftrightarrow \dfrac{az}{(z-a)^2}$

Or

$na^n u(n-1) \qquad nu(n) = nu(n-1)$

Area/Summation Property

Laplace:

$$x(t) \longleftrightarrow X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt : ROC : \text{Re}\{s\}$$

$$\int_{-\infty}^{\infty} x(t)dt = \{X(s)\}_{s=0} \rightarrow \quad \text{only whens } s=0 \textit{ is} \text{ part of ROC}$$

13. $\int_{0}^{\infty} e^{-2t}dt = ?$

Solution:

$$T = \int_{-\infty}^{\infty} \underbrace{e^{-2t}u(t)}\,dt$$

$$X(s) = \frac{1}{s+2} \qquad ROC : \text{Re}\{s\} > -2$$

$$X(s)\big|_{s=0} = \frac{1}{2}$$

14. $\int_{0}^{\infty} e^{2t}dt$

Solution:

$$I = \int_{-\infty}^{\infty} \underbrace{e^{2t}u(t)}_{X(t)}\,dt \longleftrightarrow x(s) = \frac{1}{s-2}$$

$ROC : \text{Re}\{s\} > 2$

$X(s)\big|_{s-0} \rightarrow$ Not allowed

Calculate area from time domain.

Summation property:

$$x(n) \longleftrightarrow X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n} \qquad ROC :$$

$$\sum_{n=-\infty}^{\infty} x(n) = \{X(z)\}_{z=1} \rightarrow \quad \text{only when } z=1 \text{ is part of ROC}$$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

$$X(2) = \sum_{n=-\infty}^{\infty} x(n)2^{-n} \qquad \text{when } z = 2 \text{ is part of ROC}$$

$$x(3) = \sum_{n=-\infty}^{\infty} x(n)3^{-n} \quad \text{when } z = 3 \text{ is part of ROC}$$

Calculation of summation:

$$s = \sum_{n=0}^{\infty} g(n)$$

$$s = g(0) + g(1) + g(2) + \ldots$$

G.P.

$$s = \sum_{n=-\infty}^{\infty} g(n)u(n) = \sum_{n=-\infty}^{\infty} x(n)$$

$$x(n) \longleftrightarrow x(z) \text{ ROC:}$$

$$\{X(z)\}_{z=1} : \text{if } z = 1 \text{ is part of ROC}$$

x(n) = g(n)u(n)

15. $s = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n$

Solution:

Method 1: $s = 1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \ldots\ldots\ldots = \frac{1}{1-\frac{1}{2}} = 2$

Method 2: $\sum_{n=-\infty}^{\infty} \left(\frac{1}{2}\right)^n = \sum_{n=-\infty}^{\infty} \underbrace{\left(\frac{1}{2}\right)^n u(n)}_{\downarrow}$

$$x(z) = \frac{z}{z-\frac{1}{2}} : |z| > \frac{1}{2}$$

$$x(z)\big|_{z=1} = 2$$

Test Your Understanding

3. $s = \sum_{n=0}^{\infty} a^n(n+1) \quad : \quad |a| < 1$

Solution: $x(n) \longleftarrow X(z) = \frac{az}{(z-a)^2} + \frac{z}{z-a}$

Includes unity circle $\text{ROC} : |z| > |a|$

4. $s = \sum_{n=0}^{\infty} \underbrace{\cos 2n \left(\frac{1}{2}\right)^n}$

Solution:

$$\cos 2nu(n) \longleftrightarrow \frac{z^2 - z\cos 2}{z^2 - 2z\cos 2 + 1} : |z| > 1$$

$$\downarrow \qquad \downarrow$$

$$X(z)\big|_{z=2} \qquad z = 2 \text{ is part of ROC}$$

ROC

1. BZT of x(n) may not always exist.
2. BZT of x(n) converges (finite) in a 2-D region formed by concentric circles.
3. BZT of x(n) converges in a range of r|z| value, where each r(|z|) represents a circle.
4. BZT are always associated ROC.

ROC building concept:

1. $x(n) = \delta(n) \longleftrightarrow x(z) = 1$

ROC: entire z-plane $0 \le |z| \le \infty$

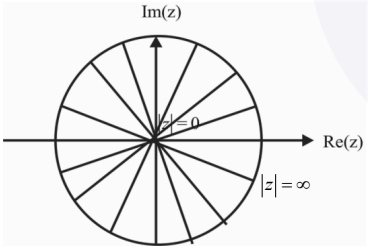


Fig. 10.8

2. $x(n) = \left(\frac{1}{2}\right)^n u(n) \longrightarrow x(z) = \frac{z}{z - \frac{1}{2}} \quad ROC : |z| > \frac{1}{2}$

$\rightarrow R.S.S$

$\rightarrow Abs\sum \quad \frac{1}{2} < |z| \le \infty$

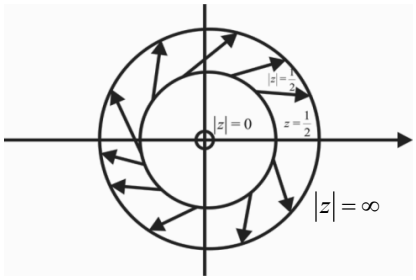


Fig. 10.9

3. $x(n) = \left(\frac{1}{2}\right)^{n-1} u(n-1) \longleftrightarrow X(z)$

$$= \left\{\frac{z}{z - \frac{1}{2}}\right\}^{z-1} \qquad |z| > \frac{1}{2}$$

$\rightarrow$ R.S.S

$\rightarrow$ Abs$\sum$ $\qquad \frac{1}{2} < |z| \le \infty$

4. $x(n) = \left(\frac{1}{2}\right)^{n+1} u(n+1) \longleftrightarrow x(z) = \frac{z^2}{z - \frac{1}{2}} : |z| > \frac{1}{2}$

R.S.S $\longrightarrow$ $\qquad$ pole : $z = \frac{1}{2}$ $\quad \frac{1}{2} < |z| < \infty$

Abs$\sum \longrightarrow$ $\qquad z = \infty$

5. $x(n) = -\left(\frac{1}{2}\right)^n u(-n-1) \longleftrightarrow x(z)$

$$= \frac{z}{z - \frac{1}{2}} \qquad |z| < \frac{1}{2}$$

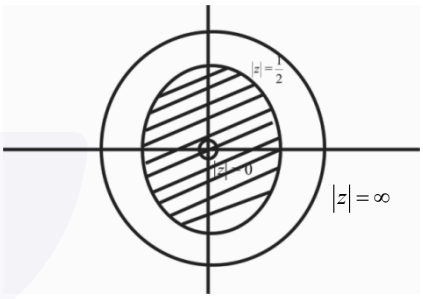


Fig. 10.10

$\rightarrow$ L.S.S.

$\rightarrow$ Abs not $\sum$ $\qquad 0 \le |z| < \frac{1}{2}$

6. $x(n) = -\left(\frac{1}{2}\right)^{n-2} u(-(n-2)-1)$

$$\longleftrightarrow \underbrace{\frac{1}{z\left(z - \frac{1}{2}\right)}} \qquad ROC : |z| < \frac{1}{2}$$

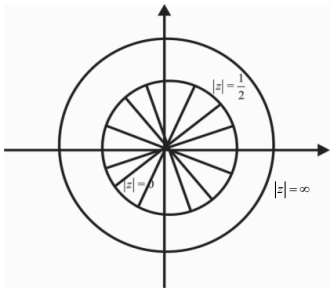


Fig. 10.11

$z = 0 \quad z = \frac{1}{2} \qquad 0 \le |z| < \frac{1}{2}$

Properties of ROC:

1. ROC of BZT may or may not include zeros of x(z).

2. ROC of BZT should be newer, including poles of x(z).
3. If x(n) is absolutely summable ROC of x(z) includes unity circle.
4. If x(n) is finite duration + Abs then ROC of x(z) is the entire z-plane except possibly and/or

5.

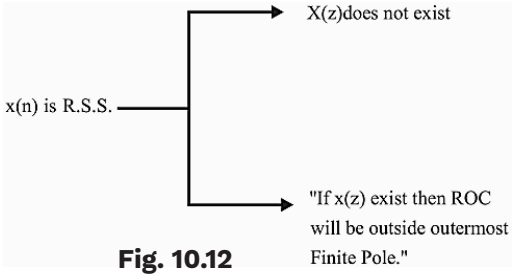


Fig. 10.12

6.

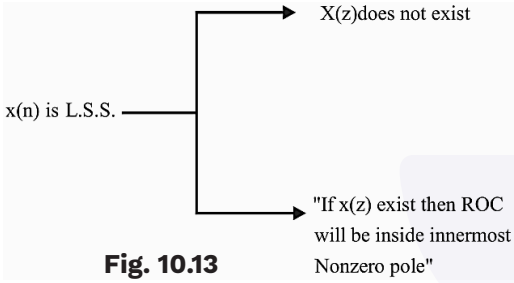


Fig. 10.13

7.

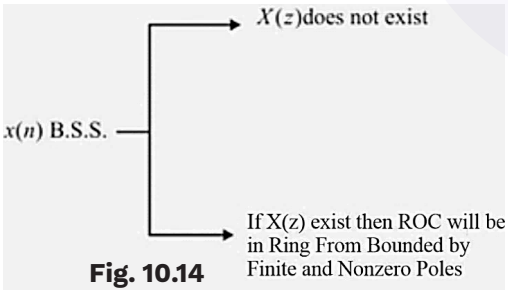
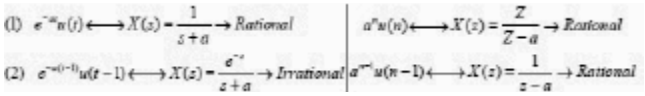


Fig. 10.14



17. $x(n)=u(n-1) \longleftrightarrow X(z)=\frac{1}{z-1}$ ROC: $|z|>1, 1\le|z|\le\infty$

Causal:

x(n)	ROC
Causal + RSS	ROC is outside outermost finite pole including $\|z\|=\infty$
Causal + {FD + Abs∑}	ROC is entire z-plane including $\|z\|=\infty$: check inclusion or removal of $\|z\|=0$

Anti-causal:

x(n)	ROC
Anti-casual + LSS	ROC is inside innermost nonzero pole including $\|z\|=0$
Anti-causal + FD + Ab∑	ROC is entire z-plane including $\|z\|=0$ and excluding $\|z\|=0$

Table 10.2

Non-causal:

x(n)	ROC
Non-causal + RSS	Outside outmost finite pole excluding $\|z\|=0$ and $\|z\|=\infty$
Non-causal + LSS	Inside innermost non zero pole excluding $\|z\|=0$ and $\|z\|=\infty$
Non-causal + (FD + Abs∑)	Entire z-plane: $0<\|z\|<\infty$ excluding $\|z\|=0$ and $\|z\|=\infty$

Table 10.3

Common mistake:

1. Before deciding the ROC first check the causality and summability, then decide the ROC accordingly.

Inverse ZT:

1. Long division → Causal $x(n)$ / Anticausal $x(n)$
2. Partial fraction
3. Residue method
4. Series expansion

"Long-division"

$$x(z)=\frac{N(z)}{D(z)}$$

S.1 $D(z) \ge N(z)$

S.2 $X(z)=\frac{N(z)}{D(z)}$ — N_1D : decreasing power of z → causal signal; N_1D : incresing power of z → anticausal signal

18. $x(z) = \dfrac{1+2z^{-1}}{1-2z^{-1}+z^{-2}}$

1. Calculate x(n) such that it is causal signal.
2. Calculate x(n) such that it is anti-causal signal.

Solution:

$$X(z) = \frac{z^2+2z}{z^2-2z+1} \longleftrightarrow x(n) = \{1, 4, 7, 10.....\}$$

Causal: retained three dots

$$Z^2 - 2z + 1 \;\overline{)\; Z^2 + 2z \;}\; 1 + 4z^{-1} + 7z^{-2} + 10z^{-3} + ...$$

$$Z^2 - 2z + 1$$

$$- \quad + \quad -$$

$$4z - 1$$

$$4z - 8 + 4z^{-1}$$

$$- \quad + \quad -$$

$$7 - 4z^{-1}$$

$$7 - 14z^{-1} + 7z^{-2}$$

$$- \quad + \quad -$$

$$10z^{-1} - 7z^{-2}$$

Anti-causal:

$$1 - 2z + z^2 \;\overline{)\; 2z + z^2 \;}\; 2z + 5z^2 + 8z^3$$

$$2z - 4z^2 + 2z^3$$

$$- \quad + \quad -$$

$$5z^2 - 2Z^3$$

$$5z^2 - 10z^3 + 5z^4$$

$$- \quad + \quad -$$

$$8z^3 - 5z^4$$

$$8z^3 - 16z^4 + 8z^5$$

$$- \quad + \quad -$$

$x(n) = \{.....8, 5, 2, 0\}$

Anti causal

retained three dots only, Anti-casual

Partial fraction:

S.1 $X(Z) = \dfrac{N(z)}{D(z)}$: Common factor cancellation.

S.2 poles: ⟶ Point

|poles| ⟶ Circles ⟶ z-plane is divided into different region ⟶ possible ROC

S.3 For each ROC then will be IZT.

19. $x(z) = \dfrac{z}{(z-1)(z-2)}$

Solution: $X(z) = \dfrac{Z}{(z-1)(z-2)}$

$z = 1 \quad z = 2$

$|z| = 1 \quad |z| = 2$

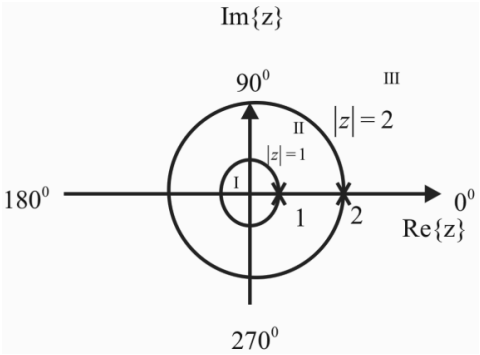


Fig. 10.15

Possible ROCs: $|z| < 1$, $1 < |z| < 2$, $|z| > 2$

$|z| < 1$

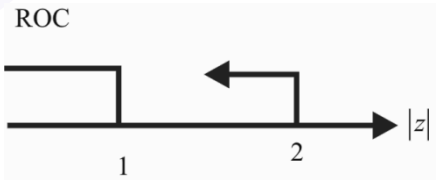


Fig. 10.16

m.1 $X(z) = \dfrac{z}{(z-1)(z-2)}$

$$\frac{X(z)}{z} = \frac{1}{(z-1)(z-2)}$$

$$\frac{X(z)}{z} = \frac{1}{z-2} - \frac{1}{z-1} \Rightarrow X(7) = \frac{z}{z-2} - \frac{z}{z-1}$$

pole $z = 2 \quad z = 1$

$|z| = 2 \quad |z| = 1$

$|z| < 2 \quad |z| < 1$

m.2 $x(n) = -(2)^n u(-n-1) + u(-n-1)$

$$X(z) = \frac{z}{(z-1)(z-2)} = \left\{ \underset{|z|<2}{\frac{2}{(z-2)}} - \underset{|z|<1}{\frac{1}{(z-1)}} \right\}$$

$$x(n) = -2(2)^{n-1} u\left(-(n-1)-1\right) + u\left(-(n-1)-1\right)$$

$$x(n) = -(2)^n u(-n) + u(-n)$$

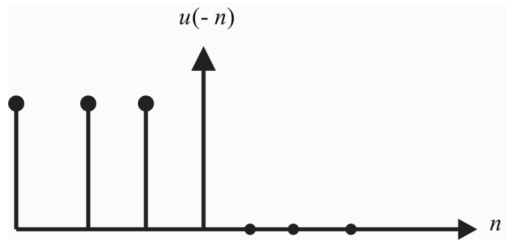


Fig. 10.17

$u[-n] = \delta(n) + u(-n-1)$

$x(n) = -(2)^n\left[\delta(n) + u(-n-1)\right] + \delta(n) + u(-n-1)$

$x(n) = -2^n\delta(n) - 2^n u(-n-1) + \delta(n) + u(-n-1)$

$x(n) = -2^n u(-n-1) + u(-n-1)$

1. $1 < |z| < 2$

$$\frac{X(z)}{z} = \frac{1}{z-2} - \frac{1}{z-1}$$

$$X(z) = \underbrace{\frac{z}{z-2}}_{|z|<2} - \underbrace{\frac{z}{z-1}}_{|z|>1}$$

$x(n) = -(2)^n u(-n-1) - u(n)$

2. $|z| > 2$:

$$X(z) = \underbrace{\frac{z}{z-2}}_{|z|>2} - \underbrace{\frac{z}{z-1}}_{|z|>1} \leftarrow x(n) = 2^n u(n) - u(n)$$

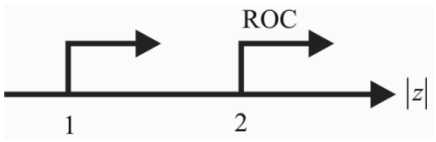


Fig. 10.18

20. $$x(z) = \frac{2 + 3z^{-1}}{(1+z^{-1})\left(1+\frac{z^{-1}}{2}\right)\left(1+\frac{z^{-1}}{4}\right)}$$

calculate x(n) = ?

Solution:

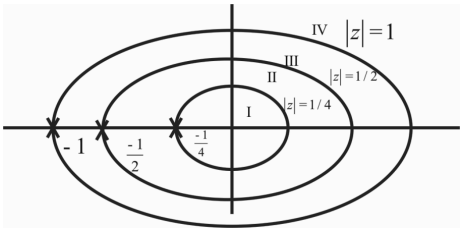


Fig. 10.19

$$X(z) = \frac{z^2(2z+3)}{(z+1)\left(z+\frac{1}{2}\right)\left(z+\frac{1}{4}\right)}$$

Pole : $z = -1 = 1\angle 180^0$

$$z = \frac{-1}{2} = \frac{1}{2}\angle 180^0$$

$$z = -\frac{1}{4} = \frac{1}{4}\angle 180^0$$

Possible ROCs:

1. $|z| < \frac{1}{4}$
2. $\frac{1}{4} < |z| < \frac{1}{2}$
3. $\frac{1}{2} < |z| < 1$
4. $|z| > 1$
5. $|z| < \frac{1}{4}$

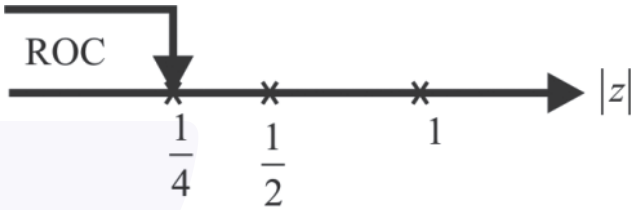


Fig. 10.20

$$\frac{X(z)}{z} = \frac{z(2z+3)}{\left(z+\frac{1}{4}\right)\left(z+\frac{1}{2}\right)(z+1)}$$

$$= -\frac{\frac{10}{3}}{\left(z+\frac{1}{4}\right)} + \frac{8}{z+\frac{1}{2}} - \frac{\frac{8}{3}}{z+1}$$

$$X(z) = \frac{-\frac{10}{3}z}{\left(z+\frac{1}{4}\right)} + \frac{8z}{\left(z+\frac{1}{2}\right)} - \frac{\frac{8}{3}z}{(z+1)}$$

$|z| = \frac{1}{4}$ $\quad |z| = \frac{1}{2}$ $\quad |z| = 1$

$|z| < \frac{1}{4}$ $\quad |z| < \frac{1}{2}$ $\quad |z| < 1$

$$x(n) = \frac{10}{3}\left(-\frac{1}{4}\right)^n u(-n-1) - 8\left(\frac{-1}{2}\right)^n$$

$$u(-n-1) + \frac{8}{3}(-1)^n u(-n-1)$$

6. $\frac{1}{4} < |z| < \frac{1}{2}$

$$X(z) = \frac{\frac{-10}{3}z}{\left(z+\frac{1}{4}\right)} + \frac{8z}{\left(z+\frac{1}{2}\right)} - \frac{-\frac{8z}{3}}{(z+1)}$$

$|z| = \frac{1}{4}$ $|z| = \frac{1}{2}$ $|z| = 1$

$|z| > \frac{1}{4}$ $|z| < \frac{1}{2}$ $|z| < 1$

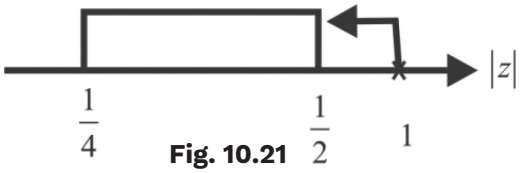


Fig. 10.21

$$x(n) = \frac{-10}{3}\left(-\frac{1}{4}\right)^n u(n) - 8\left(-\frac{1}{2}\right)^n$$

$$u(-n-1) + \frac{8}{3}(-1)^n u(-n-1)$$

7. $\frac{1}{2} < |z| < 1$

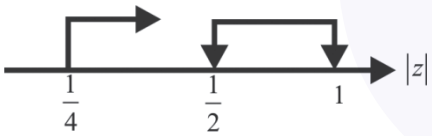


Fig. 10.22

$$X(z) = \frac{-\frac{10}{3}z}{\left(z+\frac{1}{4}\right)} + \frac{8z}{\left(z+\frac{1}{2}\right)} - \frac{\frac{8}{3}z}{(z+1)}$$

$|z| > \frac{1}{4}$ $|z| > \frac{1}{2}$ $|z| < 1$

$$x(n) = \frac{-10}{3}\left(\frac{-1}{4}\right)^n u(n) + 8\left(\frac{-1}{2}\right)^n$$

$$u(n) + \frac{8}{3}(-1)^n u(-n-1)$$

8. $|z| > 1$



Fig. 10.23

$$X(z) = \frac{\frac{-10z}{3}}{\left(z+\frac{1}{4}\right)} + \frac{8z}{\left(z+\frac{1}{4}\right)} - \frac{\frac{8z}{3}}{(z+1)}$$

$|z| > \frac{1}{4}$ $|z| > \frac{1}{2}$ $|z| > 1$

$$x(n) = \frac{-10}{3}\left(-\frac{1}{4}\right)^n u(n) + 8\left(\frac{-1}{2}\right)^n u(n) - \frac{8}{3}(-1)^n u(n)$$

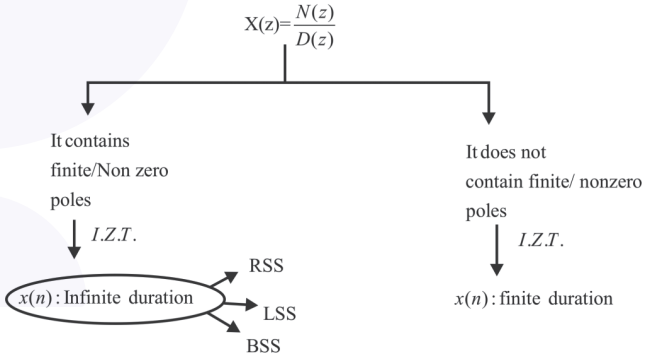


Fig. 10.24

21. Calculate x(n) such that it is non-causal.

Solution:

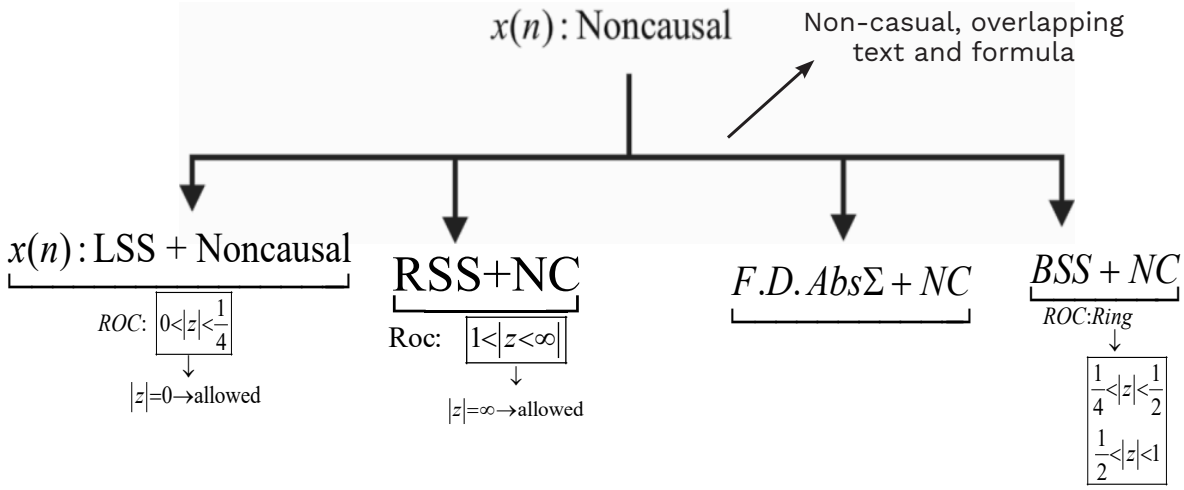


Fig. 10.25

22. $X(z) = \dfrac{z^3 + 2z^2 + 3z}{(z-1)(z-2)}$

Solution:

$$X(z) = \frac{z^3}{(z-1)(z-2)} + \frac{2z^2}{(z-1)(z-2)} + \underbrace{\left\{\frac{3z}{(z-1)(z-2)}\right\}}_{\downarrow\ X_1(z)}$$

$$\frac{X_1(z)}{z} = \frac{1}{(z-1)(z-2)} = \left\{\frac{1}{(z-2)} - \frac{1}{(z-1)}\right\}$$

$$X_1(z) = \frac{z}{(z-2)} - \frac{z}{(z-1)}$$

1. $|z| < 1 : x_1(n) = -3(2)^n u(-n-1) + 3u(-n-1)$

$$x(n) = -3(2)^n u(-n-1) + 3u(-n-1)$$
$$+$$
$$-2(2)^{n+1} u(-\overset{-n-2}{(n+1)} - 1) + 2u(-\overset{-n-2}{(n+1)} - 1)$$
$$+$$
$$-3(2)^{n+2} u(-\overset{-n-3}{(n+2)} - 1) + 2u(-\overset{-n-3}{(n+2)} - 1)$$

Possible ROCs:

(I) $|z| < 1$

(II) $1 < |z| < 2$

(III) $|z| > 2 \rightarrow$

23. $x(z) = \dfrac{z^2 + 2z}{z^2 - 2z + 4}$ Inverse ZT

Solution: Poles $\rightarrow z = 1 \pm j\sqrt{3}$ $z = 2e^{\pm j\pi/3}$ $|z| = 2$

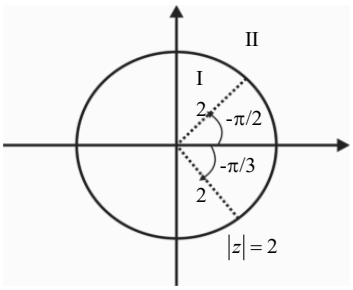


Fig. 10.26

Possible ROCs: (i) $|z| < 2$ (ii) $|z| > 2$

1. $a^n \cos\omega_0 n\, u(n) \longleftrightarrow \dfrac{z^2 - az\cos\omega_0}{z^2 - 2az\cos\omega_0 + a^2} \longleftrightarrow -a^n \cos\omega_0 n u(-n-1)$

2. $a^n \sin\omega_0 n\, u(n) \longleftrightarrow \dfrac{az\sin\omega_0}{z^2 - 2az\cos\omega_0 + a^2}$

$\longleftrightarrow -a^n \sin\omega_0 n u(-n-1)$

$$X(z) = \frac{z^2 + 2z}{z^2 - 2z + 4} = \frac{z^2 + 2z}{z^2 - 2az\cos\omega_0 + a^2}$$

$$= \frac{z^2 - z + 3z}{z^2 - 2az\cos\omega_0 + a^2}$$

$a^2 = 4 \rightarrow a = 2$

$a\cos\omega_0 = 1 \longrightarrow \omega_0 = \dfrac{\pi}{3}$

$az\cos\omega_0 = 2 \times z \times \dfrac{1}{2} = z$

$$X(z) = \frac{z^2 - az\cos\omega_0}{z^2 - 2az\cos\omega_0 + a^2} + \frac{3z(\sqrt{3})}{\sqrt{3}\left(z^2 - 2az\cos\omega_0 + a^2\right)}$$

$$X(z) = \frac{z^2 - az\cos\omega_0}{z^2 - 2az\cos\omega_0 + a^2}$$
$$+\sqrt{3}\left\{\frac{az\sin\omega_0}{z^2 - 2az\cos\omega_0 + a^2}\right\} az\sin\omega_0 = 2 \times z \times \frac{\sqrt{3}}{2} = \sqrt{3}z$$

Or

$$X(z) = \frac{z^2 + 2z}{Z^2 - 2z + 4} \rightarrow \text{complex} \qquad \begin{boxed}\begin{array}{l} z^2 - 2az\cos\omega_0 + a^2 \\ a = 2 \\ \omega_0 = \dfrac{\pi}{3}\end{array}\end{boxed}$$

$$(2)^n \cos\left(\frac{\pi}{3}n\right)u(n) \longleftrightarrow \frac{z^2 - z}{z^2 - 2z + 4}$$

$$\longleftrightarrow -(2)^n \cos\left(\frac{\pi}{3}n\right)u(-n-1)$$

$$\left(2^n\right)\sin\left(\frac{\pi}{3}n\right)u(n) \longleftrightarrow \frac{\sqrt{3}z}{z^2 - 2z + 4}$$

$$\longleftrightarrow -(2^n)\sin\left(\frac{\pi}{3}n\right)u(-n-1)$$

$$X(z) = \frac{z^2 + 2z}{z^2 - 2z + 4} = \frac{z^2 - z}{z^2 - 2z + 4} + \left\{\frac{\sqrt{3}z}{Z^2 - 2z + 4}\right\}\sqrt{3}$$

(I) $|z| < 2 : x(n) = -2^n \cos\left(\dfrac{\pi}{3}n\right)$

$$u(-n-1) - \sqrt{3}\,2^n \sin\left(\frac{\pi}{3}n\right)u(-n-1)$$

(II) $|z| > 2 : x(n) = 2^n \cos\left(\dfrac{\pi}{3}n\right)$

$$u(n) + \sqrt{3}\,(2)^n \sin\left(\frac{\pi}{3}n\right)u(n)$$

Convolution:

$x_1(n) \longleftrightarrow X_1(z) \quad R_1$

$x_2(n) \longleftrightarrow X_2(z) \quad R_2$

$x(n) = x_1(n) * x_2(n) \longleftrightarrow R_1 \cap R_2 = \{\phi\}$ Z.T. DNE

$R_1 \cap R_2 \neq \{\phi\}$

$X(z) = X_1(z)X_2(z) : R = R_1 \cap R_2$

Accumulation:

$x(n) \longleftrightarrow X(z) : \text{ ROC : R}$

$$x(n) * u(n) = \sum_{K=-\infty}^{n} x(k) \longleftrightarrow \frac{X(z)}{(1-z^{-1})}$$

if $R \cap (|z|) > 1 \neq \{\phi\}$

Difference:

$x(n) \longleftrightarrow X(z)$ ROC:R

$x(n) - x(n-1) \longleftrightarrow (1-z^{-1})X(z)$ ROC:at least R

Analogy:

$$\frac{dx(t)}{dt} \longleftrightarrow SX(S)$$

$x(n) - x(n-1) \longleftrightarrow (1-z^{-1})X(z)$

$$\int_{-\infty}^{t} x(t)dt \longleftrightarrow \frac{X(s)}{s}$$

$$\sum_{K=-\infty}^{n} x(k) \longleftrightarrow \frac{X(Z)}{1-z^{-1}}$$

Analogy:

$S \longleftrightarrow (1-z^{-1})$

Equality:

$z = e^{ST_s}$

Test Your Understanding

5. $y(n) = x(n) * x(n)$

$x(n) = (0.5)^n u(n)$

$$\sum_{n=-\infty}^{\infty} y(n) = ?$$

Solution: 4

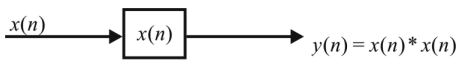
6. $y(n) = x(n) * x(n)$

$x(n) = (0.5)^n u(n)$

$$\sum_{n=-\infty}^{\infty} \left\{ \sum_{K=-\infty}^{n} y(k) \right\} = ?$$

Solution: ∞

24. $x(n) = 2^n u(-n-1)$



1. $y(n)|_{n=2} \qquad y(n)|_{n=-2}$

2. $\sum_{n=-\infty}^{\infty} y(n)$

3. $\sum_{n=-\infty}^{\infty} y(n)(-3)^{-n}$

Solution:

$$x(n) \longleftrightarrow \frac{-z}{z-2} \quad |z| < 2$$

1. $y(n) \longleftrightarrow \frac{z^2}{(z-2)^2} \quad |z| < 2$

$$-2^n u[-n-1] = \frac{z}{(z-2)}$$

$$\boxed{y(2) = -3(2)^2 u(-2) = 0}$$

$$-n2^n u[-n-1] = \frac{2z}{(z-2)^2}$$

$$\boxed{y(-2) = -(-2+1)2^{-2}u(2) = \frac{1}{4}}$$

$$y(n) = -(n+1)2^n u(-n) = \frac{z^2}{(z-2)^2} : |z| < 2$$

2. $\sum_{n=-\infty}^{\infty} y(n) = Y(z)|_{z=1} \rightarrow$ part of ROC $= \frac{1}{(1-2)^2} = 1$

3. $\sum_{n=-\infty}^{\infty} y(n)(-3)^n = Y(z)|_{z=-3} = \infty$

$\downarrow$

$z = -3 = 3\angle 180^0$

$|z| = 3 \rightarrow$ Not part of ROC

Test Your Understanding

7. x(n) = n(n+1) $\quad 0 \le n \le 4$

$y(n) = x(n) * x(n)$

$y(n)\big|_{n=4} \rightarrow$ coeff of z^{-4} ?

Solution: 84

8.

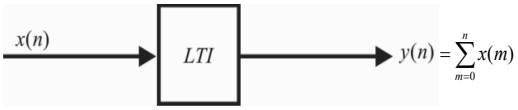


Fig. 10.27

$$y(n) \longleftrightarrow y(z) = \frac{2}{z(z-1)2}$$

$x(n)\big|_{n=4}$ = ?

Solution: 2

$x(n)$: General

$x(n)$: causal

$$\sum_{K=-\infty}^{n} x(K) \longleftrightarrow \frac{X(z)}{(1-z^{-1})}$$

$$\sum_{K=-\infty}^{n} x(K) = \sum_{K=0}^{n} x(K) \longleftrightarrow \frac{X(z)}{(1-z^{-1})}$$

DT LTI System Analysis by Using ZT

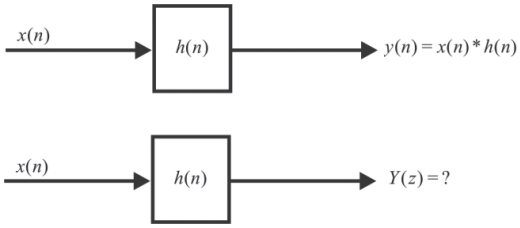


Fig. 10.28

Stability of DT LTI system:

1. For a DT LTI system to be stable h(n) must be absolutely summable.
2. For h(n) to be absolutely summable ROC of H(z) must include unity circle.
3. For DT LTI system to be stable ROC of H(z) must include unity circle.

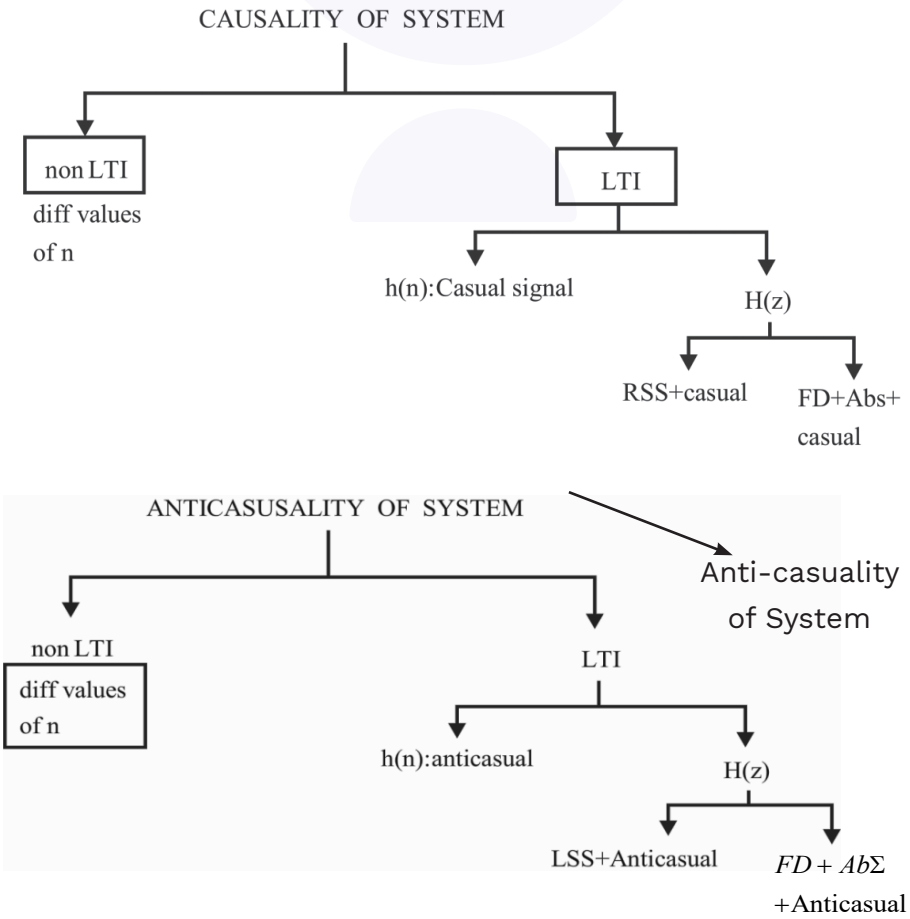


Fig. 10.29

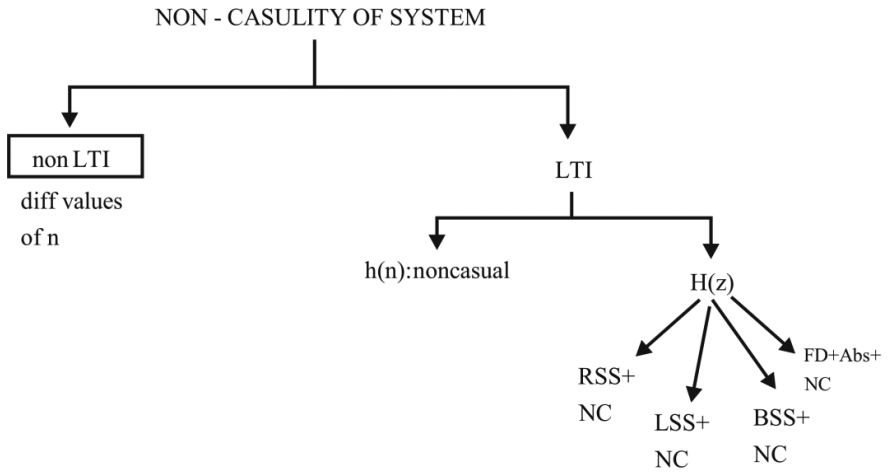


Fig. 10.30

Simultaneous causal and stable:

1. $\left[h(n):\text{causal}\right]\cap\left[\sum_{K=-\infty}^{\infty}|h(K)|:\text{Finite}\right]$

2. $h(n)=\delta(n)\longleftrightarrow H(z)=1 \quad 0\le|z|\le\infty$

$h(n)=\delta(n-1)\longleftrightarrow H(z)=\frac{1}{z} \quad 0<|z|\le\infty$

$h(n)=\left(\frac{1}{2}\right)^n u(n)\longleftrightarrow H(z)=\frac{z}{z-0.5} \quad \frac{1}{2}<|z|\le\infty$

$\underbrace{h(n)=\left(\frac{1}{2}\right)^{n+1}u(n+1)}_{\text{non causal.}}\longleftrightarrow H(z)=\frac{z^2}{z-0.5} \quad \frac{1}{2}<|z|\le\infty$

$\rightarrow H(z)=\frac{N(z)}{D(z)}$

$\{D(z)\ge N(z)\}\cap$ (All the poles of H(z) should be strictly inside unity circle)

Simultaneous anti causal and stable:

1. $\left[h(n)=\text{anti causal}\right]\cap\left[\sum_{K=-\infty}^{\infty}|h(K)|:\text{Finite}\right]$

2. $\delta[n+1]\longleftrightarrow \text{ROC}:0\le|z|<\infty$

$-2^n u[-n-1]\longleftrightarrow\frac{z}{z-2}\text{ROC}:0\le|z|<2$

$-2^n u[-(n+2)-1]\longleftrightarrow\frac{z^3}{(z-2)}\text{ROC}:0\le|z|<2$

$-2^n u[-(n-2)-1]\longleftrightarrow\frac{z^{-1}}{(z-2)}\text{ROC}:0<|z|<2$

$H(z)=\frac{N(z)}{D(z)}$

(H(z) should not include z^{-n} terms) $\cap$ (all the poles of H(z) are strictly outside unity circle)

25. $H(z)=\frac{z}{\left(z-\frac{1}{2}\right)(z-2)(z-3)}$

calculate h(n) such that LTI system is

(A) Stable

(B) Causal

(C) Non-causal

(D) Anti-causal

Solution:

S.1 Pole: $z=\frac{1}{2},2,3$

$|z|=\frac{1}{2},\ 2,\ 3$

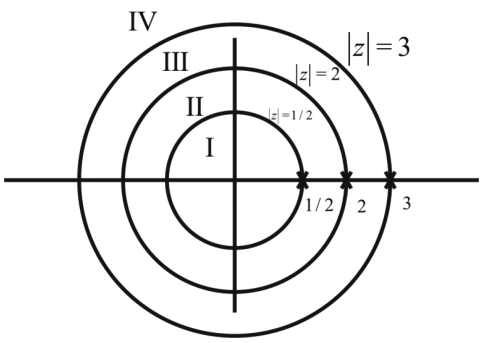


Fig. 10.31

1. $|z| < \frac{1}{2}$

2. $\frac{1}{2} < |z| < 2$ \

3. $2 < |z| < 3$

4. $|z| > 3$

a) LTI system is stable: $\frac{1}{2} < |z| < 2$

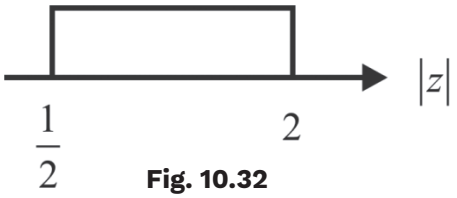


Fig. 10.32

$$\frac{H(z)}{z} = \frac{1}{\left(z-\frac{1}{2}\right)(z-2)(z-3)}$$

$$= \frac{4}{15\left(z-\frac{1}{2}\right)} - \frac{2}{3(z-2)} + \frac{2}{5(z-3)}$$

$$H(z) = \frac{4}{15}\frac{z}{z-\frac{1}{2}} - \frac{2z}{3(z-2)} + \frac{2z}{5(z-3)}$$

$|z| > \frac{1}{2}$ $|z| < 2$ $|z| < 3$

$$h(n) = \frac{4}{15}\left(\frac{1}{2}\right)^n u(n) + \frac{2}{3}2^n u(-n-1) - \frac{2}{5}3^n u(-n-1)$$

2.

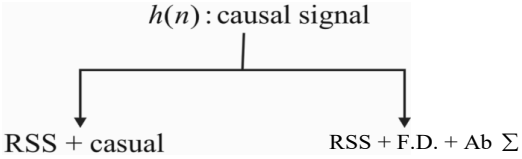


Fig. 10.33

Req^n : $3 < |z| \le \infty$
Check : $z = \infty \rightarrow$ Finite

$3 < |z| \le \infty \rightarrow$ causal signal

$3 < |z| \le \infty$

$$H(z) = \frac{\frac{4}{15}z}{\left(z-\frac{1}{2}\right)} - \frac{\frac{2}{3}z}{(z-2)} + \frac{\frac{2z}{5}}{(z-3)}$$

$|z| > \frac{1}{2}$ $|z| > 2$ $|z| > 3$

$$h(n) = \frac{4}{15}\left(\frac{1}{2}\right)^n u(n) - \frac{2}{3}(2)^n u(n) + \frac{2}{5}3^n u(n)$$

3.

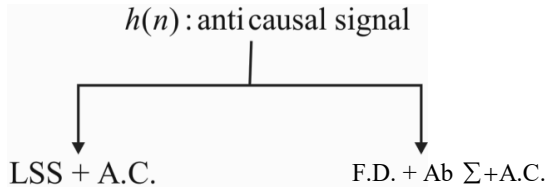


Fig. 10.34

Req^n : $0 \le |z| < \frac{1}{2}$
Check : $z = 0 \rightarrow$ Finite

$0 \le |z| < \frac{1}{2} \longrightarrow$ A.C. h(n)

$|z| < \frac{1}{2}$

$$H(z) = \frac{\frac{4}{15}z}{\left(z-\frac{1}{2}\right)} - \frac{\frac{2}{3}z}{(z-2)} + \frac{\frac{2}{5}z}{(z-3)}$$

$$h(n) = \frac{-4}{15}\left(\frac{1}{2}\right)^n u(-n-1) + \frac{2}{3}(2)^n u(-n-1) - \frac{2}{3}(3)^n u(-n-1)$$

4.

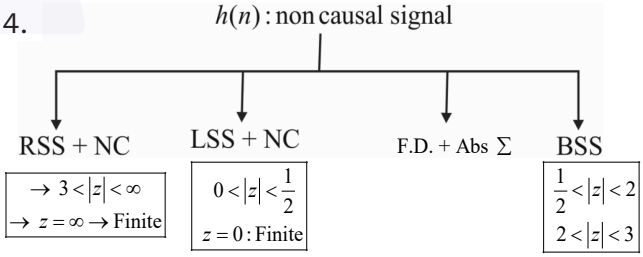


Fig. 10.35

Unilateral ZT:

$x(n) \xleftrightarrow{BLT} x(z)$ $x(n) \xleftrightarrow{ULT} x(z)$

$$x(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n} \qquad x(z) = \sum_{n=0}^{\infty} x(n)z^{-n}$$

$ULT[x(n)] = BLT[x(n)u(n)]$

1. $1 \xleftrightarrow{BLT}$ D.N.E.

$1 \xleftrightarrow{ULT}$

$$ULT[1] = ULT[1.u(n)] = \frac{z}{z-1} \qquad ROC : |z| > 1$$

$1 \longleftrightarrow \frac{z}{z-1}$

$a^n \longleftrightarrow \frac{z}{z-a}$

$\cos \omega_0 n \longleftrightarrow \frac{z^2 - z\cos\omega_0}{z^2 - 2z\cos\omega_0 + 1}$

$\sin \omega_0 n \longleftrightarrow \frac{z\sin\omega_0}{z^2 - 2z\cos\omega_0 + 1}$

Time shifting:

BLT	ULT
$x(n) \longleftrightarrow x(z)$	$x(n) \longleftrightarrow x(z)$
$x(n-1) \longleftrightarrow z^{-1}x(z)$	$x(n-1) \longleftrightarrow z^{-1}x(z) + x(-1)$
$x(n-2) \longleftrightarrow z^{-2}x(z)$	$x(n-2) \longleftrightarrow z^{-2}x(z) + z^{-1}x(-1) + x(-2)$
$x(n-3) \longleftrightarrow z^{-3}x(z)$	$x(n-3) \longleftrightarrow z^{-3}x(z) + z^{-2}x(-1) + z^{-1}x(-2) + x(-3)$
$x(n) \longleftrightarrow x(z)$	$x(n) \longleftrightarrow x(z)$
$x(n+1) \longleftrightarrow zx(z)$	$x(n+1) \longleftrightarrow zx(z) - zx(0)$
$x(n+2) \longleftrightarrow z^2x(z)$	$x(n+2) \longleftrightarrow z^2x(z) - z^2x(0) - zx(1)$

Table 10.4

Linear constant coefficient difference equation (LCCDE):

1. $y(n+2) + y(n-1) + 2y(n) = 3x(n) \longrightarrow n \to n-2$
2. $y(n-2) + 2y(n-1) + y(n) = 3x(n)$

Types of response:

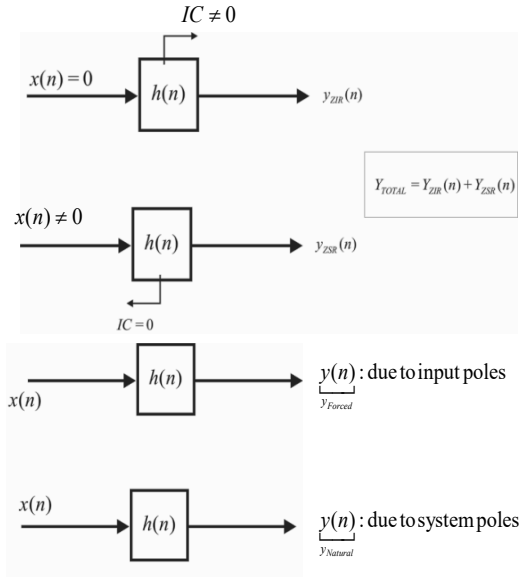


Fig. 10.36

$$\boxed{Y_{Total} = Y_{Forced} + Y_{Natural}}$$

Transfer function to DE and vice versa:

Case 1: IC = 0

1. $H(z) = \frac{Y(z)}{X(z)}$ 1: TF can be calculated.
2. Y(z) can be calculated from TF

$$Y(z) = X(z)H(z)$$

$$\updownarrow$$

$$y(n) = ?$$

Case 2: IC ≠ 0

$\to H(z) = \left\{\frac{Y(z)}{X(z)}\right\}_{IC=0}$ can be calculated.

$\to$ Y(z) can't be calculated from TF Y(z) = ?

S.1 From DES.2 Apply proper

S.3 Regenerate IC

26. $y(n+2) - \frac{5}{6}y(n+1) + \frac{1}{6}y(n) = 5x(n+1) - x(n)$

With IC: $y(-1) = 2$, $y(-2) = 0$ and $x(n) = u(n)$

(A) Calculate ZIR/ZSR

$$x(z) = \frac{1}{(1-z^{-1})} \quad \boxed{\begin{matrix} x(-1) = 0 \\ x(-2) = 0 \end{matrix}}$$

(B) Calculate forced / Natural.

Solution: $n \to n-2$

$$y(n)-\frac{5}{6}y(n-1)+\frac{1}{6}y(n-2)=5x(n-1)-x(n-2)$$

$$Y(z)-\frac{5}{6}[z^{-1}Y(z)+y(-1)]$$

$$+\frac{1}{6}\left[z^{-2}Y(z)+z^{-1}y(-1)+y(-2)\right]$$

$$=5[z^{-1}X(z)+x(-1)]-[z^{-2}X(z)+z^{-1}x(-1)+x(-2)]$$

$$Y(z)-\frac{5}{6}[z^{-1}Y(z)+2]+\frac{1}{6}[z^{-2}Y(z)+z^{-1}2]$$

$$=[5z^{-1}-z^{-2}]X(z)$$

$$\boxed{Y(z)=\left\{\frac{\frac{5}{3}-z^{-\frac{1}{3}}}{(1-\frac{5}{6}z^{-1}+\frac{1}{6}z^{-1})}\right\}+\left\{\frac{5z^{-1}-z^{-2}}{(1-\frac{5}{6}z^{-1}+\frac{1}{6}z^{-2})}\right\}X(z)}$$

$$Y(z)=\underbrace{\left\{\frac{3}{\left(1-\frac{1}{2}z^{-1}\right)}-\frac{\frac{4}{3}}{\left(1-\frac{1}{3}z^{-1}\right)}\right\}}_{ZIR}$$

$$+\underbrace{\left\{\frac{12}{(1-z^{-1})}-\frac{18}{\left(1-\frac{z^{-1}}{2}\right)}+\frac{6}{\left(1-\frac{1}{3}z^{-1}\right)}\right\}}_{ZSR}$$

$$y(n)=\underbrace{\left\{3\left(\frac{1}{2}\right)^n-\frac{4}{3}\left(\frac{1}{3}\right)^n\right\}}_{ZIR}+\underbrace{\left\{12-18\left(\frac{1}{2}\right)^n+6\left(\frac{1}{3}\right)^n\right\}}_{ZSR}:n\geq -2$$

$$y(n)=\left\{\frac{12}{(1-z^{-1})}\right\}+\left\{\frac{-15}{(1-\frac{1}{2}z^{-1})}+\frac{\frac{14}{3}}{(1-\frac{1}{3}z^{-1})}\right\}$$

↓ Forced ↓ Natural

$$y(n)=12+\left\{-15\left(\frac{1}{2}\right)^n+\frac{14}{3}\left(\frac{1}{3}\right)^n\right\}n\geq -2$$

↓ Forced ↓ Natural

Initial value theorem:

(1) $x(n)$: must be causal $\qquad z=e^{sT_s}: z\to\infty$

(2) $X(z)=\frac{N(z)}{D(z)}: \boxed{D(z)\geq N(z)} \qquad s\leftarrow(1-z^{-1})$

$$\lim_{n\to 0}x(0)=\lim_{z\to\infty}\left[\left(1-z^{-1}\right)X(z)\right] \qquad \lim_{t\to 0^+}x(0^+)=\lim_{s\to\infty}sX(s)$$

$$\boxed{x(0)=\lim_{z-\infty}X(z)}$$

$D(z)\geq N(z)$

Final value theorem:

$$\lim_{n\to\infty}x(n)=\lim_{z\to 1}(1-z^{-1})X(z) \qquad z=e^{sT_s}$$

$s\longleftrightarrow(1-z^{-1})$

$$x(\infty)=\lim_{s\to 0}sX(s)$$

1. $x(n)$: casual
2. $(1-z^{-1})X(z)$ should have all poles strictly inside unity circle.

Relation between ZT and LT theorems to avoid mistake:

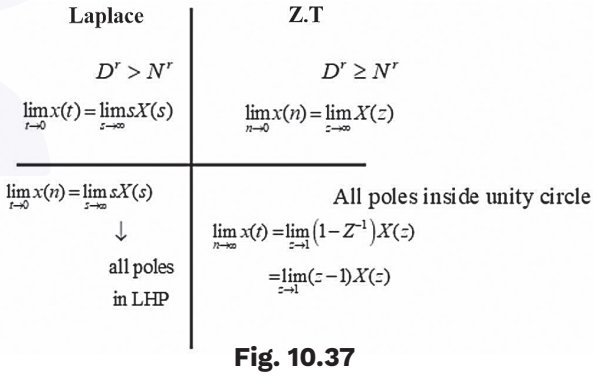


Fig. 10.37

Conjugate property:

$x(n)\longleftrightarrow x(z) \qquad ROC: |z|$

$x^*(n)\longleftrightarrow X^*(z^*) \quad ROC|z^*|$

Case 1:

$x(n)$: Real : $x(n)=x^*(n)$

$X(z)=X^*(z^*)$

If $z=z_0$(POLE/ZERO) $\longrightarrow z=z_0^*$(POLE/ZERO)

Case 2:

$x(n)$: Real + EVEN

$x(n)=x^*(n)=x(-n)$

$X(z) = X^*(z^*) = x(z^{-1})$

Let $z = 2e^{j30^0}$ is pole

$z^* = 2e^{-j30^0}$

$\frac{1}{z} = 0.5e^{-j30^0}$

$\frac{1}{z^*} = 0.5e^{j30^0}$

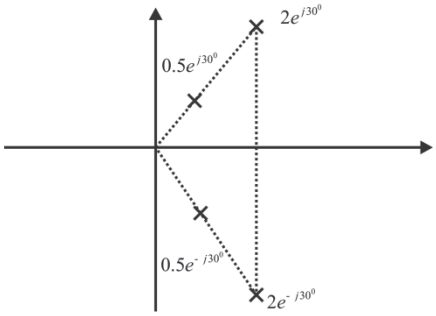


Fig. 10.38

Case 3:

x(n) : Real+odd

$x(n) = x^*(n) = -x(-n)$

$X(z) = X^*(z^*) = -X(z^{-1})$

$z = z_0 S$ pole

$z = z_0^*, \; z = \frac{1}{z_0}, z = \frac{1}{z_0^*}$ poles

ILT and IZT involving log:

$x(t) \longleftrightarrow X(s)$	$x(n) \longleftrightarrow X(z)$
$tx(t) \longleftrightarrow \frac{-dx(s)}{ds}$	$nx(n) \longleftrightarrow \frac{-zdx(z)}{dz}$

27. $X(s) = \ln\left\{\frac{s^2 + a^2}{s^2 + b^2}\right\}$

ROC: Re{s}>0 ILT =?

Solution:

$X(s) = \ln(s^2 + a^2) - \ln(s^2 + b^2)$

$tx(t) \longleftrightarrow \frac{-dX(s)}{ds}$

$tx(t) \longleftrightarrow -\left[\frac{2s}{s^2 + a^2} - \frac{2s}{s^2 + b^2}\right] = \frac{2s}{s^2 + b^2} - \frac{2s}{s^2 + a^2}$

$tx(t) = [2\cos bt\, u(t) - \cos at\, u(t)]$

$x(t) = \frac{2}{t}(\cos bt\, u(t) - \cos at\, u(t))$

Block Diagram Representation

1. A linear system can be characterised by its transfer function H(z).
2. Figure (a) shows a block diagram of a system with a transfer function H(z) and its input and output X(z), and Y(z), respectively.
3. Subsystems may be interconnected by using cascade (series), parallel, and feedback interconnections, the three elementary types.

Cascade interconnection:

1. When the systems are connected in series, as depicted in fig. (b), then the transfer function of the overall system is the product of the individual transfer function.
2. This result can also be proved by observing that in fig.(b):

$$H(z) = \frac{Y(z)}{X(z)} = \frac{W(z)}{X(z)}\frac{Y(z)}{W(z)} = H_1(z)H_2(z)$$

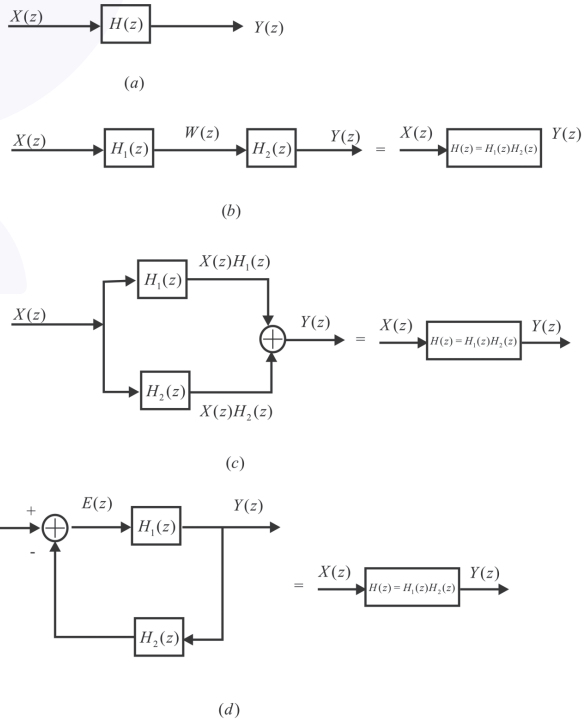


Fig. 10.39 Elementary Connection of Block Diagrams

Parallel interconnection:

1. When the systems are connected in parallel, as depicted in fig.(c), then the transfer function of the overall system is the sum of the individual transfer function.

2. This result can also be proved by observing that in fig.(c):

$$Y(z) = X(z)H_1(z) + X(z)H_2(z)$$

$$Y(z) = X(z)[H_1(z) + H_2(z)]$$

$$\frac{Y(z)}{X(z)} = H(z) = H_1(z) + H_2(z)$$

We can extend this result to any number of systems connected in parallel.

Feedback interconnection:

1. When the output is fed back to the input, as shown in fig. (d), the overall transfer function $H(z) = Y(z) / X(z)$ can be computed as follows.
2. The input to the adder are $X(z)$ and $-H_2(z)Y(z)$. Therefore, $E(z)$, the output of the adder, is

$$E(z) = X(z) - H_2(z)Y(z)$$

But

$$Y(z) = H_1(z)E(z)$$

$$= H_1(z)[X(z) - H_2(z)Y(z)]$$

$$Y(z) = H_1(z)X(z) - H_1(z)H_2(z)Y(z)$$

$$Y(z) = [1 + H_1(z)H_2(z)] = H_1(z)X(z)$$

$$\frac{Y(z)}{X(z)} = H(z) = \frac{H_1(z)}{1 + H_1(z)H_2(z)} \quad \ldots(i)$$

Therefore the feedback loop can be replaced by a single block with the transfer function shown in eq. (i). The transfer function $H_1(z)$ is the forward path transfer function, and $H_2(z)$ is the feedback path transfer function.

System Realisation

1. We now develop a systematic method for realisation (or implementation) of an arbitrary *N*th-order transfer function. The most general *N*th-order transfer function is given by

$$H(z) = \frac{b_0 z^N + b_1 z^{N-1} + ... + b_{N-1} z + b_N}{z^N + a_1 z^{N-1} + ... + a_{N-1} z + a_N}$$

2. A given transfer function can be realised in many different ways.
3. A transfer function $H(z)$ can be conveniently represented in a block diagram from using the basic building block representing the unit delay, the multiplier, the adder, and the pick-off node as shown in fig.

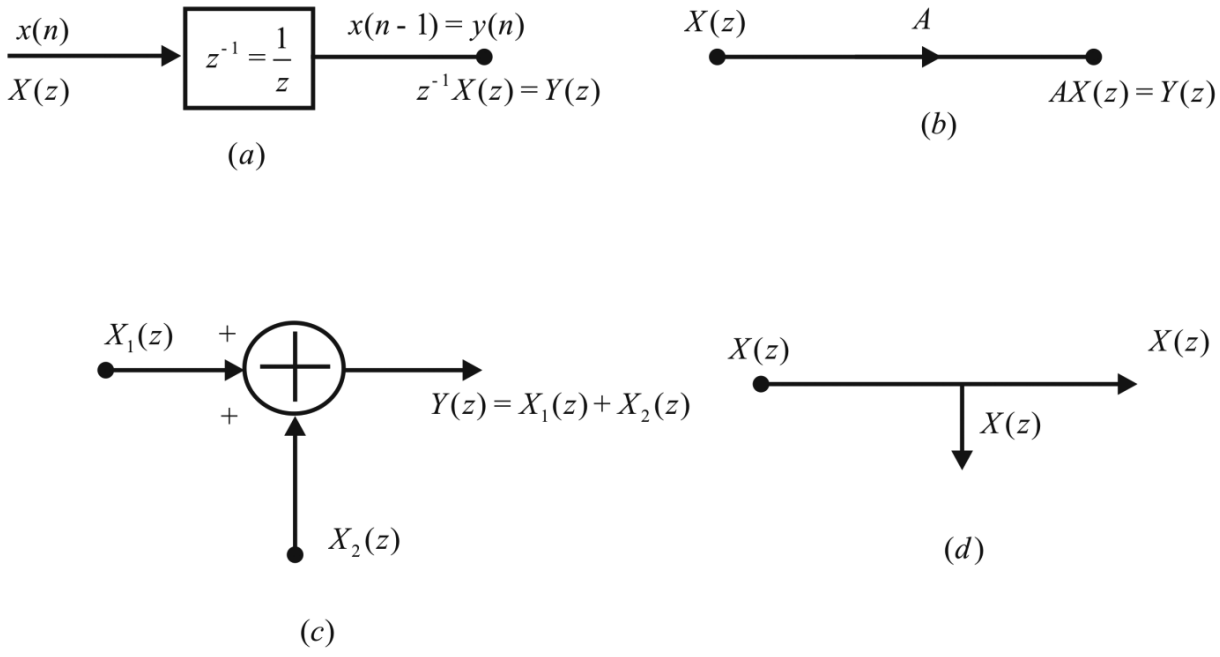


Fig. 10.40

Fig. 10.2 Basic Building Blocks Representing (a) Unit-Delay, (b) Multiplier, (c) Adder and (d) Pick-Off

Direct form I realisation:

1. An *N*th-order transfer function is characterised by 2N + 1 unique coefficients and, in general, requires 2N + 1 multipliers and 2N two-input adders for implementation.
2. Structures in which the multipliers coefficients are precisely the coefficients of the transfer function are called direct from structures.

   Consider for simplicity a third-order system characterised by a transfer function:

$$H(z) = \frac{b_0 z^3 + b_1 z^2 + b_2 z + b_3}{z^3 + a_1 z^2 + a_2 z + a_3}$$

$$H(z) = \frac{b_0 + (b_1 / z) + (b_2 / z^2) + (b_3 / z^3)}{1 + (a_1 / z) + (a_2 / z^2) + (a_3 / z^3)}$$

We can express H(z) as

$$H(z) = \underbrace{\left(b_0 + \frac{b_1}{z} + \frac{b_2}{z^2} + \frac{b_3}{z^3}\right)}_{H_1(z)} \underbrace{\left(\frac{1}{1 + (a_1 / z) + (a_2 / z^2) + (a_3 / z^3)}\right)}_{H_2(z)}$$

We can realise H(z) as a cascade of transfer function $H_1(z)$ followed by $H_2(z)$, as depicted in fig. (a), where

$$H_1(z) = \frac{W(z)}{X(z)} = b_0 + \frac{b_1}{z} + \frac{b_2}{z^2} + \frac{b_3}{z^3}$$

$$W(z) = \left(b_0 + \frac{b_1}{z} + \frac{b_2}{z^2} + \frac{b_3}{z^3}\right) X(z)$$

$$H_2(z) = \frac{Y(z)}{W(z)} = \frac{1}{1 + (a_1 / z) + (a_2 / z^2) + (a_3 / z^3)}$$

$$Y(z) = W(z) - \left(\frac{a_1}{z} + \frac{a_2}{z^2} + \frac{a_3}{z^3}\right) Y(z)$$

The left-half section of fig.(b) shows the realisation of $H_1(z)$.

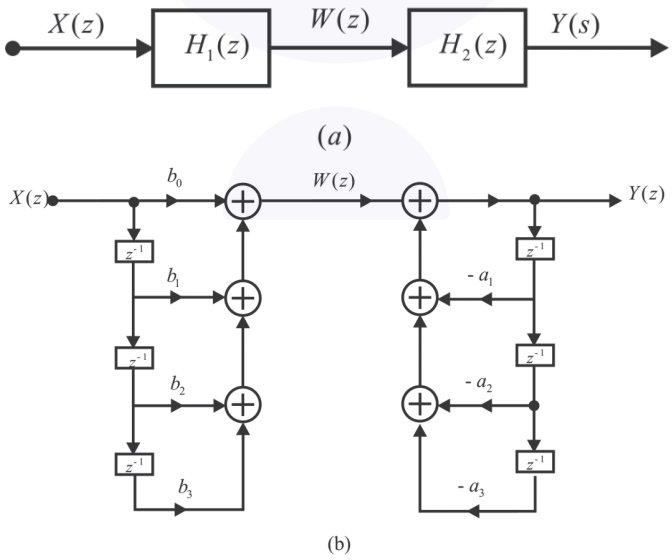


Fig. 10.41 (a) Realisation of a Transfer Function in Two Steps and (b) Direct form-I Realisation of a Third-Order Discrete-Time LTI System

a) We can generalise this procedure, known as the direct from I realisation, for any value of N.
b) This procedure requires 2 *N* delay units to realise an *N*th-order transfer function.
c) This realisation is noncanonic since it employs six delay units to implement a third-order transfer function.
d) A realisation is canonic if the number of delay units used in the realisation is equal to the order of the transfer function realised.
e) Thus, canonic realisation has no redundant delay units.

Direct form II (or Canonic) realisation:

1. In the direct from I, we realise $H(z)$ by implementing $H_1(z)$ followed by $H_2(z)$, as shown in fig.(a).
2. We can also realise $H(z)$, as shown in fig. (a), where $H_2(z)$ is followed by $H_1(z)$.
3. This procedure is known as the direct form II realisation.
4. The direct form I realisation [Fig.(b)] implements zeros first [the left-half section represented by $H_2(z)$] followed by realisation of poles [the right-half section represented by $H_2(z)$]of $H(z)$.
5. In contrast, the direct form II realisation implements poles first, followed by zeros. Again consider for simplicity a third-order system characterised by a transfer function:

$$H(z) = \frac{b_0 z^3 + b_1 z^2 + b_2 z + b_3}{z^3 + a_1 z^2 + a_2 z + a_3}$$

$$H(z) = \frac{b_0 + (b_1 / z) + (b_2 / z^2) + (b_3 / z^3)}{1 + (a_1 / z) + (a_2 / z^2) + (a_3 / z^3)}$$

We can express $H(z)$ as

$$H(z) = \underbrace{\left(\frac{1}{1 + (a_1 / z) + (a_2 / z^2) + (a_3 / z^3)}\right)}_{H_2(z)} \underbrace{\left(b_0 + \frac{b_1}{z} + \frac{b_2}{z^2} + \frac{b_3}{z^3}\right)}_{H_1(z)}$$

We can realise $H(z)$ as a cascade of transfer function $H_2(z)$ followed by $H_1(z)$, as depicted in fig. (a), where

$$H_2(z) = \frac{V(z)}{X(z)} = \frac{1}{1 + (a_1 / z) + (a_2 / z^2) + (a_3 / z^3)}$$

$$V(z) = X(z) - \left(\frac{a_1}{z} + \frac{a_2}{z^2} + \frac{a_3}{z^3}\right) V(z)$$

and

$$H_1(z) = \frac{Y(z)}{V(z)} = b_0 + \frac{b_1}{z} + \frac{b_2}{z^2} + \frac{b_3}{z^3}$$

$$Y(z) = \left(b_0 + \frac{b_1}{z} + \frac{b_2}{z^2} + \frac{b_3}{z^3}\right) V(z)$$

6. The left-half section of fig.(b) shows the realisation of $H_2(z)$ and the right-half section shows the realisation of $H_1(z)$.
7. This implementation halves the number of delay units to N, and thus more efficient in hardware utilisation.
8. This is the direct form II realisation. This realisation is canonic since it employs N delay units to implement a *N*th-order transfer function.

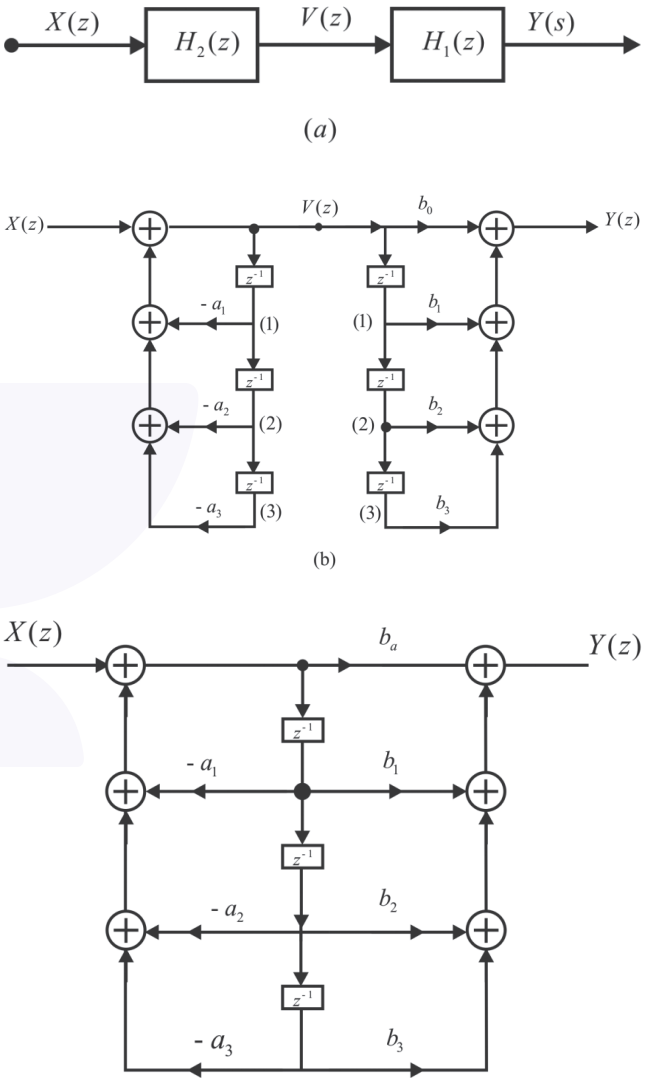


Fig. 10.42 Direct Form II Realisation of a Third-Order LTI System

10.1. Find the canonic (direct form II) realisation of

$$H(z) = \frac{1 - \left(\frac{7}{4}\right) z^{-1} - \left(\frac{1}{2}\right) z^{-2}}{1 + \left(\frac{1}{4}\right) z^{-1} - \left(\frac{1}{8}\right) z^{-2}}$$

Solution:

Given that

$$H(z)=\frac{Y(z)}{X(z)}=\frac{V(z)}{X(z)}\frac{Y(z)}{V(z)}=\frac{1-(7/4)z^{-1}-(1/2)z^{-2}}{1+(1/4)z^{-1}-(1/8)z^{-2}}$$

$$=\frac{1-(7/4)(1/z)-(1/2)(1/z^2)}{1+(1/4)(1/z)-(1/8)(1/z^2)}$$

$$=\underbrace{\left(\frac{1}{1+(1/4)(1/z)-(1/8)(1/z^2)}\right)}_{H_2(z)}$$

$$\underbrace{(1-(7/4)(1/z)-(1/2)(1/z^2))}_{H_1(z)}$$

Where

$$H_2(z)=\frac{V(z)}{X(z)}=\frac{1}{1+(1/4)(1/z)-(1/8)(1/z^2)}$$

$$V(z)=X(z)-\frac{1}{4}\frac{V(z)}{z}+\frac{1}{8}\frac{V(z)}{z^2}$$

And

$$H_1(z)=\frac{Y(z)}{V(z)}=1-\frac{7}{4}\frac{1}{z}-\frac{1}{2}\frac{1}{z^2}$$

$$Y(z)=V(z)-\frac{7}{4}\frac{V(z)}{z}-\frac{1}{2}\frac{V(z)}{z^2}$$

The given transfer function $H(z)=(1-7/4z^{-1}-1/2z^{-2})/(1+1/4z^{-1}-1/8z^{-2})$ is of the second-order; therefore, we need only two delay units for its realisation as shown in fig.

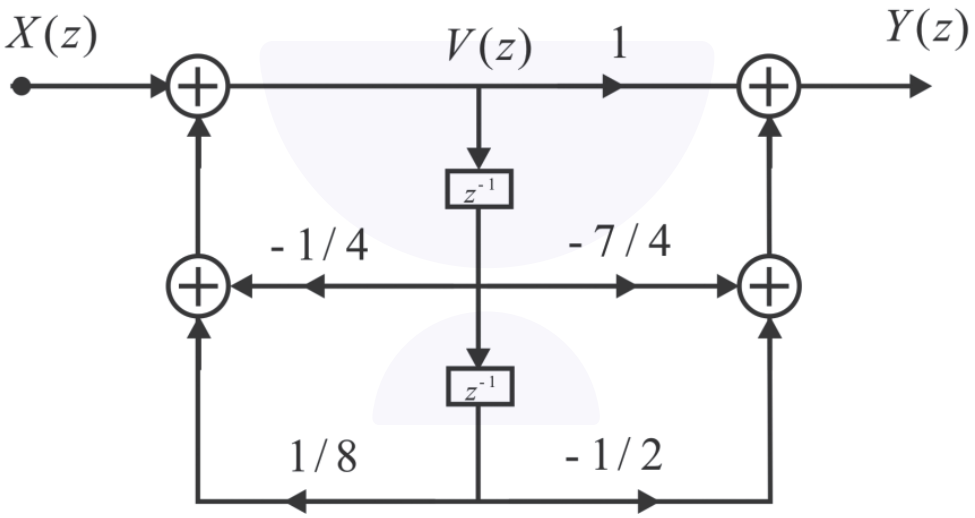


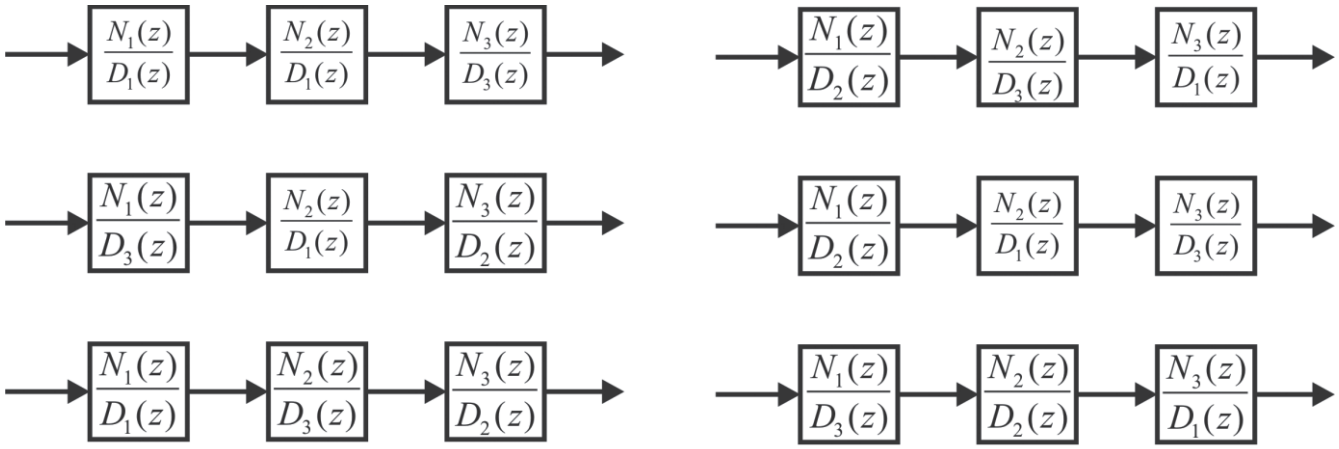
Fig. 10.43 Direct Form II Realisation of H(z) of Example 10.65

Cascade and parallel realisation:

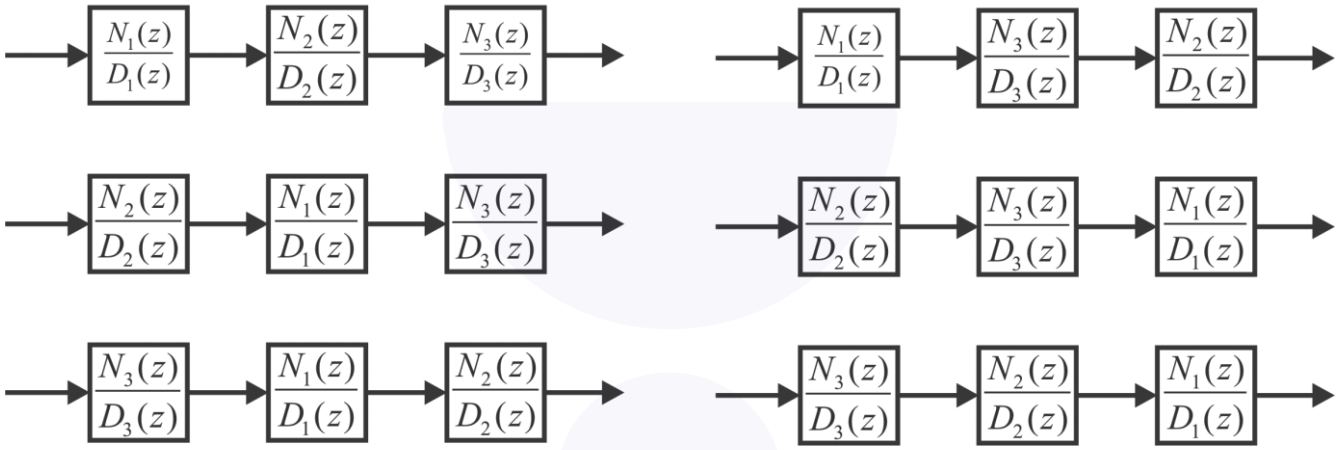
1. By expressing the numerator polynomial $N(z)$ and the denominator polynomial $D(z)$ of the transfer function, $H(z)$ as a product of polynomials of lower degree is often realised as a cascade of low-order systems.
2. Consider, for example, $H(z)=N(z)/D(z)$ expressed as

$$H(z)=\frac{N_1(z)N_2(z)N_3(z)}{D_1(z)D_2(z)D_3(z)}$$

3. Various different cascade realisations of $H(z)$ can be obtained by different pole-zero polynomial pairings.
4. Some examples of such realisations are shown in fig(a).
5. Additional cascade realisations are obtained by simply changing the ordering of the systems.
6. Figure(b) illustrates examples of different structures obtained by changing the ordering of the systems.



(a)



(b)

Fig. 10.44 Different Cascade Realisations Obtained by (a) Different Pole-Zero Pairing and (b) Changing the Ordering of the Systems

Transposed realisation:

1. Two realisations are said to be equivalent if they have the same transfer function.
2. A simple way to generate an equivalent structure from a given realisation is via the transpose operation, which is as follows:
   a) Reverse all paths
   b) Replace pick-off nodes with adders, and vice versa.
   c) Interchange the input and output nodes.

   Figure shows the transposed version of the canonic direct from realisation.

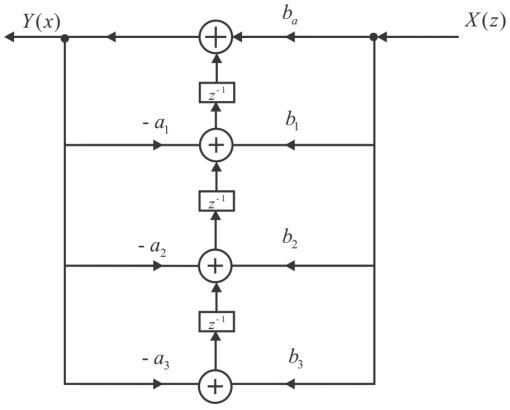


Fig. 10.45 Transposed Version of the Canonic Direct Form II Realisation

10.2. Find the canonic(direct form-II) and transposed realisation of the following transfer function:

1. $H(z) = \dfrac{4z + 28}{z + 1}$

2. $H(z) = \dfrac{4z + 28}{z^2 + 6z + 5}$

Solution:

1. Given that

$$H(z) = \frac{Y(z)}{X(z)} = \frac{V(z)}{X(z)}\frac{Y(z)}{V(z)} = \frac{4z + 28}{z + 1}$$

$$= \frac{4 + 28/z}{1 + 1/z} = \underbrace{\frac{1}{1 + 1/z}}_{H_2(z)}\underbrace{4 + \frac{28}{z}}_{H_1(z)}$$

Where

$$H_2(z) = \frac{V(z)}{X(z)} = \frac{1}{1 + 1/z}$$

$$V(z) = X(z) - \frac{V(z)}{z}$$

And

$$H_2(z) = \frac{Y(z)}{V(z)} = 4 + \frac{28}{z}$$

$$Y(z) = 4V(z) + 28\frac{V(z)}{z}S$$

a) The given transfer function $H(z) = (4z + 28)/(z + 1)$ is of the first-order; therefore, we need only one delay unit for its realisation.

b) Figure (a) shows the canonic (direct from II), and fig. (b) its transpose.

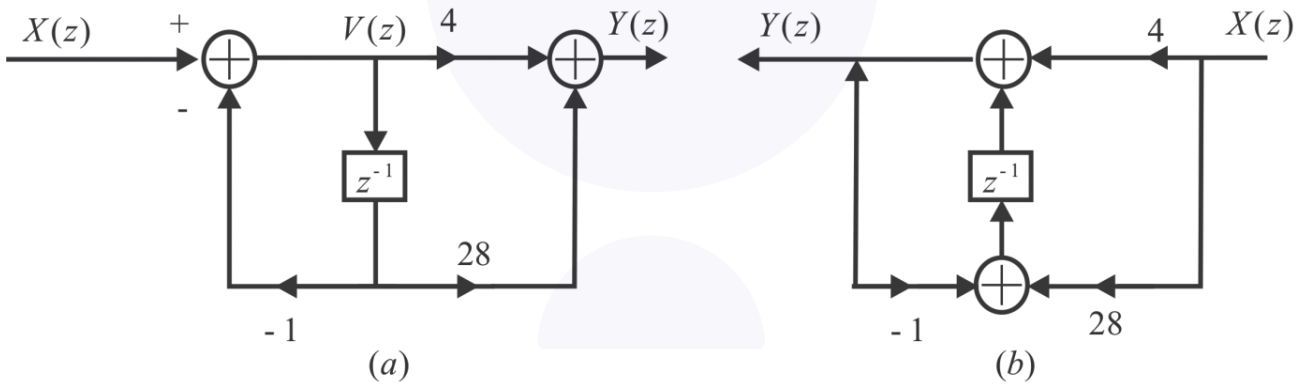


Fig. 10.46

2. Given that

$$H(z) = \frac{Y(z)}{X(z)} = \frac{V(z)}{X(z)}\frac{Y(z)}{V(z)} = \frac{4z + 28}{z^2 + 6z + 5} = \frac{(4/z) + (28/z^2)}{1 + (6/z) + (5/z^2)} = \underbrace{\left(\frac{1}{1 + (6/z) + (5/z^2)}\right)}_{H_2(z)}\underbrace{\left(\frac{4}{z} + \frac{28}{z^2}\right)}_{H_1(z)}$$

Where

$$H_2(z) = \frac{V(z)}{X(z)} = \frac{1}{1 + (6/z) + (5/z^2)}, \quad V(z) = X(z) - 6\frac{V(z)}{z} - 5\frac{V(z)}{z^2}$$

And

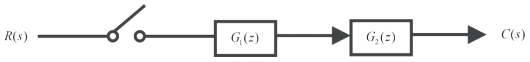
$$H_1(z) = \frac{Y(z)}{V(z)} = \frac{4}{z} + \frac{28}{z^2}$$

$$Y(z) = 4\frac{V(z)}{z} + 28\frac{V(z)}{z^2}$$

The given transfer function $H(z) = (4z + 28)/(z^2 + 6z + 5)$ is of the second-order; therefore, we need only two delay units for its realisation.

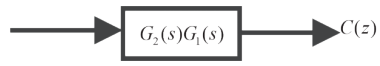
Exercise Level 1

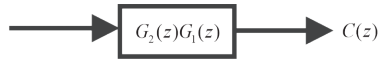
1. For the given sampled-data system

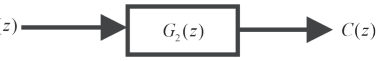


The z-transform is

(A) 

(B) 

(C) 

(D) 

2. Match list-I (Evaluation of the value of function) with list-II (corresponding z-transform expression) and select the correct answer using the code given below the lists:

List-I	**List-II**
A. Final value	1. $\lim_{z\to 0}(1-z^{-1})F(z)$
B Initial value	2. $\lim_{z\to 1}(1-z^{-1})F(z)$
	3. $\lim_{z\to \infty}F(z)$
	4. $\lim_{z\to \infty}zF(z)$

Codes:

	A	**B**
(A)	1	3
(B)	1	4
(C)	2	3
(D)	2	4

3. Let x(n)be a left-sided sequence that is equal to zero for $n > 0$. If $X(z) = \dfrac{3z^{-1} + 2z^{-2}}{3 - z^{-1} + z^{-2}}$ then x(0) will be

(A) 0
(B) 2
(C) 3
(D) 4

4. The partial fraction expansion of the function is

$$R(z) = \frac{4z^2 - 2z}{z^3 - 5z^2 + 8z - 4}$$

(A) $\dfrac{2}{z-1} + \dfrac{12}{(z-2)^2}$

(B) $\dfrac{2}{z-1} + \dfrac{2}{z-2} + \dfrac{12}{(z-2)^2}$

(C) $\dfrac{1.5}{z-1} + \dfrac{12}{(z-1)(z-2)}$

(D) $\dfrac{1.5}{z-1} + \dfrac{1.5}{z-2} + \dfrac{1}{(z-2)^2}$

5. A discrete-time LTI system with rational system function H(z) is causal if and only if

(A) The ROC is the exterior of a circle outside the outermost pole.
(B) The ROC is the interior of a circle outside the outermost pole.
(C) The ROC is the exterior of a circle outside the innermost pole.
(D) The ROC is the interior of a circle outside the innermost pole.

Exercise Level 2

1. For a z-transform

$$X(z) = \frac{z\left(2z - \frac{5}{6}\right)}{\left(z - \frac{1}{2}\right)\left(z - \frac{1}{3}\right)}$$

Match list-I (the sequences) with list-II (the region of convergence) and select the correct answer using the codes given below the lists:

List-I

(A) $[(1/2)^n + (1/3)^n]u(n)$

(B) $(1/2)^n u(n) - (1/3)^n u(-n-1)$

(C) $-(1/2)^n u(-n-1) + (1/3)^n u(n)$

(D) $-[(1/2)^n + (1/3)^n]u(-n-1)$

List-II

1. $(1/3) < |z| < (1/2)$
2. $|z| < (1/3)$
3. $|z| < 1/3$ & $|z| > 1/2$
4. $|z| > 1/2$

Codes:

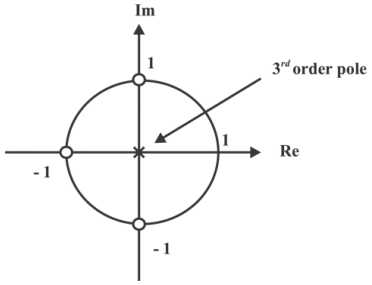
	A	B	C	D
(A)	4	2	1	3
(B)	1	3	4	2
(C)	4	3	1	2
(D)	1	2	4	3

2. The transfer function of a stable discrete-time LTI system is $H(z) = \dfrac{K(z-a)}{(z+0.5)}$ where K and α are real numbers. The value of α (rounded off to one decimal place) with $|\alpha| > 1$, for which magnitude response of the system is constant over all frequencies, is _____.

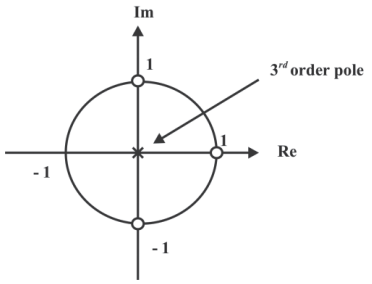
3. Which one of the following pole-zero plots corresponds to the transfer function of an LTI system characterised by the input-output difference equation given below?

$$y[n] = \sum_{k=0}^{3} (-1)^k x[n-k]$$

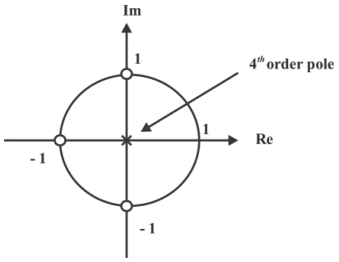
(A)



(B)



(C)



(D)

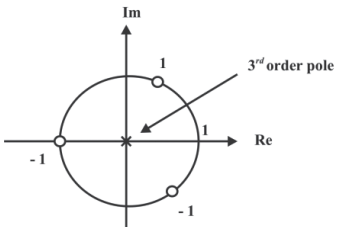
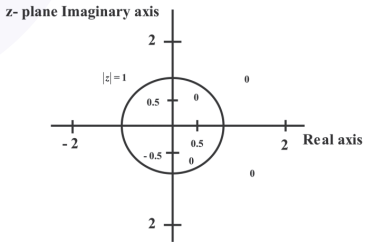


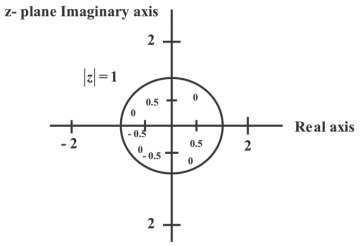
Fig. 10.47

4. Let H(z) be the z-transform of a real-valued discrete-time signal h[n]. If $P(z) = H(z)H\left(\dfrac{1}{z}\right)$ has a zero at $z = \dfrac{1}{2} + \dfrac{1}{2}j$, and P(z) has a total of four zeros, which one of the following plots represents all the zeros correctly?

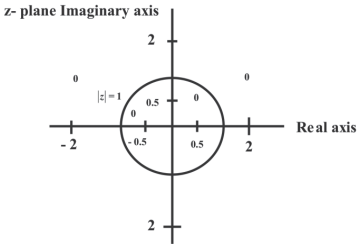
(A)



(B)



(C)



(D)

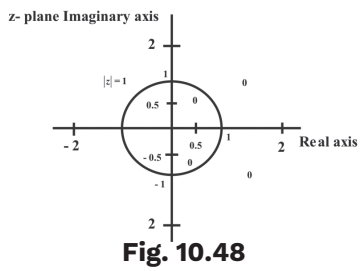
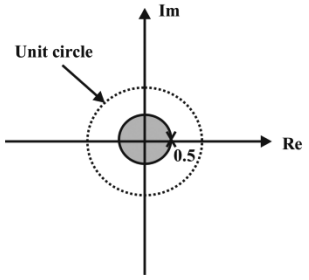


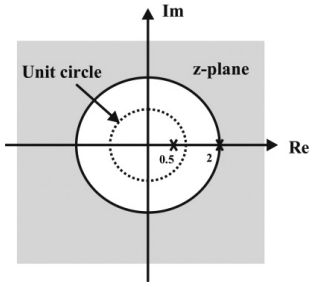
Fig. 10.48

5. The ROC (region of convergence) of the z-transform of a discrete-time signal is represented by the shaded region in the z-plane. If the signal $x[n] = (2.0)^{|n|}$, $-\infty < n < +\infty$, then the ROC of its z-transform is represented by

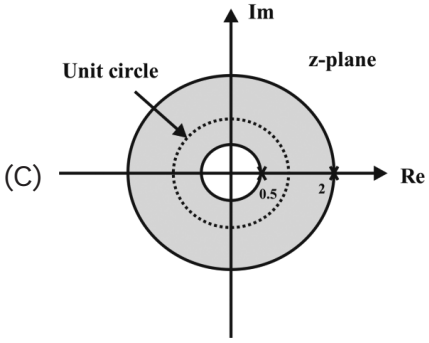
(A)



(B)



(C)



(D)

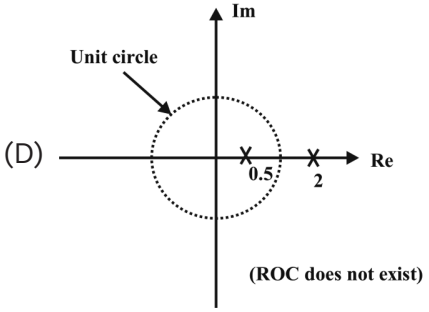


Fig. 10.49

Answer Key

Level 1									
1.	**(C)**	**2.**	**(C)**	**3.**	**(B)**	**4.**	**(B)**	**5.**	**(A)**
Level 2									
1.	**(C)**	**2.**	**-2**	**3.**	**(B)**	**4.**	**(D)**	**5.**	**(D)**

Solution Level - 1

1. Solution: (C)

2. Solution: (C)
As per the definition of initial value and final value theorem.

Solution Level - 2

1. Solution: (C)

$$X(z) = \frac{z\left(2z - \frac{5}{6}\right)}{\left(z - \frac{1}{2}\right)\left(z - \frac{1}{3}\right)} = \frac{z}{z - \frac{1}{2}} + \frac{z}{z - \frac{1}{3}}$$

We know that

$$X(z) = \frac{z}{z - a}$$

$$\Rightarrow x(n) = a^n u(n), |z| > |a|$$

$$or\ x(n) = -a^n u(-n-1), |z| < |a|$$

If $|z| > \frac{1}{2}$,

then $x(n) = \left\{\left(\frac{1}{2}\right)^n + \left(\frac{1}{3}\right)^n\right\} u(n)$

if $\frac{1}{3} < |z| < \frac{1}{2}$

then $x(n) = -\left(\frac{1}{2}\right)^n u(-n-1) + \left(\frac{1}{3}\right)^n u(n)$

if $|z| < \frac{1}{3}$

then $x(n) = -\left(\frac{1}{2}\right)^n u(-n-1) - \left(\frac{1}{3}\right)^n u(-n-1)$

If $|z| < \frac{1}{3} \& |z| > \frac{1}{2}$,

then $x(n) = \left(\frac{1}{2}\right)^n u(n) - \left(\frac{1}{3}\right)^n u(-n-1)$

2. Solution: -2

Given transfer function for discrete-time LTI system

$$H(z) = \frac{K(z-a)}{(z+0.5)}$$

Where 'K' and 'α' are real numbers.

As it is asked to find the value of 'α' such that the magnitude of response is constant over all frequencies, so we have to find the value of 'α' for which the given LTI system will behave as an all-pass filter.

Location of poles and zeros follows the reciprocal conjugate relationship as poles and zeros are symmetrical about the unit circle.

$$p = \frac{1}{z^*}$$

where {P= location of pole, Z= location of zero

For given system

$p = -0.5 \qquad z = \alpha$

$$\therefore -0.5 = \frac{1}{\alpha^*}$$

Given, 'α' is real, so, $\alpha^* = \alpha$

$$-0.5 = \frac{1}{\alpha} \qquad \therefore \alpha = -2$$

Hence, the value of α is -2.

3. Solution: (B)

Method 1

$$y[n] = \sum_{k=0}^{3} (-1)^k \delta[n-k]$$

To find impulse response, substitution $x[n] = \delta[n]$ in the given input-output relation.

$$h[n] = \sum_{k=0}^{3} (-1)^k \delta[n-k]$$

$$h[n] = \delta[n] - \delta[n-1] + \delta[n-2] - \delta[n-3]$$

Taking z-transform both sides to find the transfer function of the system,

$$H(z) = 1 - z^{-1} + z^{-2} - z^{-3}$$

$$H(z) = \frac{z^3 - z^2 + z - 1}{z^3}$$

$$H(z)\Big|_{z=+1} = \frac{1-1+1-1}{1} = 0$$

So, the system function has a zero at z=+1, that indicates that the low-frequency gain of the filter in zero.

So it will be a high pass filter. To check, finding

$H(z)$ at $\omega = \pi$ or $z = -1$

$$H(z)\Big|_{z=-1} = \frac{-1-1-1-1}{-1} = 4$$

So the system behaves as an HPF, and it will not have zero at z=-1.

The only option satisfying this is an option (B).

Hence, the correct option is (B).